

Endocrine Aspects of Chronic Kidney Disease

Juan Jesús Carrero | Peter Stenvinkel | Bengt Lindholm

CHAPTER OUTLINE

PANCREATIC HORMONAL DISORDERS:
INSULIN RESISTANCE, 1901
HYPOTHALAMOPITUITARY AXIS, 1904

PROLACTIN, 1908
VITAMIN D, PARATHYROID HORMONE, AND
KIDNEY DISEASE, 1912

The kidney is a potent endocrine organ, a key modulator of endocrine function, and an important target for hormonal action. Thus alterations in signal feedback mechanisms and in production, transport, metabolism, elimination, and protein binding of hormones occur rather commonly in conditions affecting the kidney. As a direct consequence, chronic kidney disease (CKD), end-stage kidney disease (ESKD), and kidney transplantation are all associated with abnormalities in the synthesis or action of many hormones. The purpose of this chapter is to present an overview of specific endocrine abnormalities that manifest as a consequence of kidney disease.

PANCREATIC HORMONAL DISORDERS: INSULIN RESISTANCE

Insulin resistance (IR) is a common feature of CKD—regardless of underlying cause—that describes a clinical condition in which there is a reduced biologic effect for any given blood concentration of insulin.¹ The body's resistance to the actions of insulin results in a compensatory increase in the production and secretion of insulin by the pancreas and leads to hyperinsulinemia to maintain euglycemia. If there is a concomitant inadequate secretion of insulin, this condition is manifested as abnormal glucose tolerance and, if severe, as diabetes mellitus. In CKD, both hyperparathyroidism and vitamin D deficiency may mediate insulin secretory abnormalities. Indeed, treatment of hyperparathyroidism² and pharmacologic doses of vitamin D³ have been reported to correct glucose tolerance. The gold standard for determining IR is the euglycemic hyperinsulinemic clamp method.⁴ However, because the clamp method is complex, expensive, and impractical to perform in large population studies, several surrogate methods (e.g., the homeostasis model assessment [HOMA]) have been developed. Unfortunately, all surrogate methods of determining insulin resistance suffer from important limitations, including poor precision, and oral glucose tolerance test (OGTT)–derived insulin sensitivity indices may be preferred to fasting sample–derived indices.⁵ Insulin

resistance is seen as the common denominator for a number of metabolic disturbances, including hyperinsulinemia, impaired glucose tolerance, fatty liver, abdominal obesity, hyperuricemia, elevated triglyceride levels, low high-density-lipoprotein (HDL) cholesterol levels, and hypertension.⁶

CAUSES OF UREMIC INSULIN RESISTANCE

The seminal study by DeFronzo and colleagues has demonstrated that impaired tissue sensitivity to insulin is the primary cause of insulin resistance in uremia, implying a postreceptor defect in peripheral skeletal muscle.⁷ Still, the molecular site of the postreceptor defect in humans has not been clearly established.⁸ A recent study using the method of metabolic phenotyping and controls has confirmed that uremic patients exhibit IR with a compensatory increase in insulin secretion.⁹ A reduced level of insulin receptor phosphorylation mediated by inflammation has been observed, both in animal models of diabetes and in patients with type 2 diabetes.¹⁰ Whether the same mechanism is operative in CKD is unknown. Studies in animal models of uremia have reported downregulation of GLUT-4 in muscle,¹¹ failure of insulin to activate pyruvate dehydrogenase,¹² downregulation of insulin receptor substrate 1 (IRS-1)–associated phosphoinositol 3-kinase (PI3K), and upregulation of IRS-2–associated PI3K activity.¹³ Recent studies based on hyperinsulinemic-euglycemic-euaminoacidemic clamp and ²³Na–magnetic resonance imaging (to measure the concentration of Na⁺ in the skin and muscle) in hemodialysis (HD) patients and controls have indicated that excessive muscle Na⁺ content might also be a determinant of IR in HD patients.¹⁵ Thus in the uremic milieu, the cause of IR is multifactorial, and many metabolic alterations may contribute concurrently (Table 56.1).

One feature of advanced CKD is low levels of 1,25-dihydroxyvitamin D (1,25[OH]₂D), a pleiotropic vitamin with multiple noncalcemic functions. Evidence of an important role for vitamin D in uremic IR was provided in an international study of patients undergoing dialysis in whom 1,25(OH)₂D infusion corrected glucose intolerance and IR

Table 56.1 Multiple Causes and Consequences of Insulin Resistance and Abnormalities of Insulin Secretion in Chronic Kidney Disease

Causes	Consequences
Insulin secretion abnormalities:	Dyslipidemia
Hyperparathyroidism	Sodium retention
Vitamin D deficiency	Vascular calcification
Insulin resistance:	Muscle wasting
Uremic toxins	Hyperuricemia
Anemia	Renin activation
Metabolic acidosis	Hypertension and
Inflammation	cardiovascular disease
Oxidative stress	
Muscle loss and excessive muscle sodium	
Increased fat mass	
Physical inactivity	

in the absence of changes in parathyroid hormone (PTH).¹⁶ Although many studies in the general population have demonstrated that low serum 25-hydroxyvitamin D concentrations are associated with IR and diabetes risk,¹⁷ the precise mechanism(s) whereby vitamin D supplementation improves IR is unknown. Metabolic acidosis is another common complication in uremia, also associated with IR, and 2 weeks of oral sodium bicarbonate supplementation in patients undergoing dialysis have been found to improve insulin sensitivity.¹⁶ As in the general population without kidney disease, fat mass seems to be an important risk factor for IR in those with CKD.¹⁸ Considering the obesity epidemic, the possible role of high fructose intake in the development of metabolic syndrome, and possibly CKD, definitely needs to be highlighted. Fructose may not only induce metabolic syndrome, hyperuricemia, and weight gain¹⁹ but also exerts direct adverse effects on renal tubular cells.²⁰

Persistent inflammation and protein-energy wasting (PEW) are two frequent and interrelated features of the uremic milieu that also may mediate IR. Loss of skeletal muscle mass results in abnormal glucose disposal, so it seems logical that treatment of PEW with intravenous nutrition improved IR in a study of surgical patients.²¹ Elevated circulating proinflammatory cytokine values may also mediate IR.²² Indeed, metabolic syndrome seems to be associated with inflammatory markers²³ and leukocyte count²⁴ in CKD. Among inflammatory mediators, tumor necrosis factor (TNF; partially produced by fat tissue) especially influences the ability of insulin to stimulate glucose transport.²⁵ Blockade of interleukin-1 (IL-1) with anakinra (an IL-1 receptor antagonist) in patients with type 2 diabetes is associated not only with less inflammation but also with improved glycemic control.²⁶ It has been shown that the suppressors of cytokine signaling (SOCS) family of proteins not only exacerbate IR but also inhibit insulin signaling and insulin-like growth factor (IGF) signaling.¹⁴ On the basis of these findings, and from the fact that SOCS-1 knockout mice have a low blood glucose level and increased insulin signaling,¹⁴ the SOCS proteins have been suggested to represent an important link between elevations of cytokine levels and IR. In patients with type 2 diabetes, high IL-6 levels are associated with increased SOCS-3 expression in skeletal

muscle, and IL-6-induced SOCS-3 expression inhibits insulin signaling in human differentiated myotubes grown in vitro.²⁷ In CKD, few studies have yet examined the links between inflammation and IR. However, an inflammatory response during HD is linked to elevated SOCS-3 values and IR.²⁸

In uremic mice, urea-induced generation of reactive oxygen species (ROS) has been suggested to induce IR,²⁹ although the mechanism thereof remains to be proven in humans. Additional factors that may contribute to uremic IR include physical inactivity and anemia. Correction of anemia by erythroid-stimulating agents (ESAs) has been reported to reverse IR in patients undergoing HD, independently of iron overload.³⁰ Furthermore, a small study in nonobese, nondiabetic, stable patients undergoing dialysis has demonstrated the beneficial effect of ESA treatment on IR.³¹ The observation that IR is associated with reduced responsiveness to ESA in patients undergoing dialysis may, in part, be attributed to the presence of persistent inflammation in both conditions.³² Finally, IR in patients receiving peritoneal dialysis (PD) merits discussion because the dialysis procedure itself, in addition to the uremic state, appears to modulate the magnitude of the IR.³³ IR may be exacerbated by the intraperitoneal presence of glucose-containing dialysate.^{34,35} PD has significantly higher IR than HD, which may reflect the presence of glucose-based dialysate in the peritoneal cavity.^{36,37}

INSULIN RESISTANCE AS A RISK FACTOR FOR CHRONIC KIDNEY DISEASE

Whether IR (and metabolic syndrome) is an antecedent of CKD or merely a consequence of impaired kidney function has been debated. Large population-based studies have shown that the metabolic syndrome is associated with an increased risk for incident CKD.^{38,39} A dose-response relationship has been documented between the number of metabolic syndrome traits and the prevalence of microalbuminuria.³⁹ Furthermore, a smaller scale study of nondiabetic patients with CKD has shown that IR estimated by HOMA is 2.5 times more prevalent in patients with CKD than in controls. Finally, a Japanese study in nondiabetic patients with hypertension and stage 3 CKD has shown that IR (estimated by both HOMA and the IR index) is a significant risk factor for the deterioration of kidney function.⁴⁰ The cross-sectional design of these studies precludes causal inferences, and mechanistic and intervention studies are needed to resolve whether the metabolic syndrome is a cause, consequence, or both of CKD. However, insulin has potent growth-stimulating properties and may have proliferative effects on glomerular and mesangial cells, and it also stimulates transforming growth factor- β (TGF- β) and the renin-angiotensin-aldosterone system (RAAS). Although insulin may also promote fibrosis,⁴¹ it is not clear whether therapy directed to tackle the metabolic syndrome will delay, or even halt, the progression of CKD.

INSULIN RESISTANCE AND CARDIOVASCULAR RISK

In a prospective observational study of nondiabetic patients with ESKD, IR was associated with cardiovascular mortality (Fig. 56.1) independently of Quetelet's index (body mass index, BMI), hypertension, and dyslipidemia.⁴² Thus as in the general population, IR and ensuing hyperinsulinemia

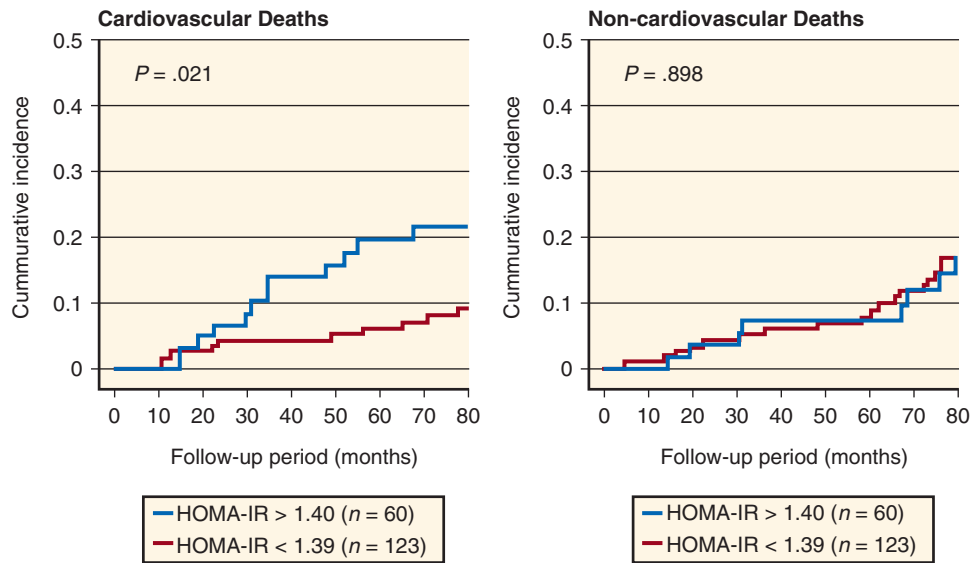


Fig. 56.1 Kaplan-Meier curves showing the association between the homeostasis model assessment (HOMA)-estimated insulin resistance (HOMA-IR) and mortality in 183 nondiabetic patients with end-stage kidney disease (ESKD) treated with maintenance hemodialysis. (Modified from Shinohara K, Shoji T, Emoto M, et al: Insulin resistance as an independent predictor of cardiovascular mortality in patients with end-stage renal disease. *J Am Soc Nephrol.* 2002;13:1894-1900.)

may be independent risk factors for cardiovascular complications in patients with ESKD. A recent study based on the CRIC (Chronic Renal Insufficiency Cohort) cohort with 6.3 years of follow-up has shown that in addition to anemia, inflammation, and poor glycemic control, IR also was an independent predictor of congestive heart failure in CKD.⁴³

Hyperinsulinemia promotes tubular sodium retention,⁴⁴ decreases urinary uric acid clearance, and upregulates the RAAS,⁴⁵ all established risk factors for hypertension. Another atherogenic link between IR and CVD is uremic dyslipidemia, typically characterized by hypertriglyceridemia, reduced HDL cholesterol, increased very low-density-lipoprotein (VLDL), and small dense low-density-lipoprotein (LDL) particles. Of note, uremic lipid abnormalities improve concomitantly with the correction of IR and glucose intolerance after the intravenous administration of 1,25(OH)₂D therapy.¹⁶

Because insulin accelerates calcium deposition in human vascular smooth muscle cells,⁴⁶ it has been speculated that although the beneficial metabolic and vasomotor effects of insulin are impaired in IR, the mitogenic signals are enhanced, and the resulting imbalance can promote vascular calcification.⁴⁷ Finally, the role of insulin in protein turnover merits attention, particularly given that altered protein turnover may have a role in CKD, cardiovascular disease (CVD), and PEW. Animal models of insulin deficiency have suggested that the effect of insulin on protein turnover is mediated through the activation of the ubiquitin proteasome pathway.⁴⁸ In accordance, uremic patients with type 2 diabetes who are undergoing dialysis have increased skeletal muscle protein breakdown in comparison with their nondiabetic counterparts.⁴⁹

TREATMENT OF INSULIN RESISTANCE IN CHRONIC KIDNEY DISEASE

Management of IR in uremia should be multifaceted. In addition to attention to and treatment of the many uremic

metabolic alterations that may lead to IR and/or impaired insulin secretion, other more specific treatment options should be discussed with the patient. Regular exercise and lifestyle modification should be an integral part of the management of IR. Whereas a small interventional trial in patients receiving HD did not show IR improvements after 3 months of aerobic exercise training,⁵⁰ a study has shown that a 12-week lifestyle modification program in 295 diabetic and nondiabetic Japanese patients reduced albuminuria, maintained their estimated glomerular filtration rate (eGFR) and reduced fasting glucose levels in patients with the metabolic syndrome.⁵¹ Because angiotensin-converting enzyme (ACE) inhibitor treatment appears to improve insulin sensitivity and may reduce the risk of type 2 diabetes mellitus in patients with essential hypertension,⁵² the effect of ACE inhibitors or angiotensin receptor blockers (ARBs) on IR needs to be tested in CKD patients. The Diabetes Reduction Assessment with ramipril and rosiglitazone Medication (DREAM) trial,⁵³ involving participants without CVD but with impaired levels of fasting glucose, has shown that ramipril does not reduce the incidence of diabetes or death but that it does facilitate the regression of glucose level elevations to normoglycemia.

In the evaluation of new users of oral hypoglycemic medication monotherapy in patients with type 2 diabetes mellitus, a higher risk of mortality was associated with glibenclamide, glipizide, and rosiglitazone than with metformin.⁵⁴ Because rosiglitazone was found to reduce the incidence of type 2 diabetes mellitus substantially and also increase the likelihood of regression to normoglycemia,⁵⁵ thiazolidinedione (glitazone) treatment was thought to be an attractive treatment option for patients with CKD. However, subsequent studies have provided conflicting results. A cross-sectional evaluation has shown significantly higher cardiovascular and all-cause mortality in rosiglitazone users,⁵⁶ consistent with a systematic review of trials in patients with type 2 diabetes mellitus that showed an increased risk of myocardial infarction and a

borderline increased risk of death from cardiovascular causes.⁵⁷ In any case, thiazolidinedione treatment in ESKD cannot currently be advocated until its efficacy has been demonstrated in randomized controlled trials. A later randomized clinical trial in patients with CKD has shown that short-term rosiglitazone therapy reduces IR but has no effect on arterial function and stiffness.⁵⁸

The insulin sensitizer metformin is a first-line pharmacologic agent for obesity and type 2 diabetes that has been widely used for over 5 decades. It is associated with a reduction in hemoglobin A1c (HbA1c) and improvements in micro- and macrovascular complications. Of note, this pleiotropic biguanide also has interesting additional outcomes, such as anticancer and life extension effects.⁵⁹ However, metformin is cleared by the kidneys, so there is a risk of metformin accumulation and associated lactic acidosis in CKD. Clinical practice guidelines have suggested that the drug should be used with caution when the estimated glomerular filtration rate (eGFR) is less than 60 mL/min/1.73 m², and the advisory board of European Renal Best Practice and the European Medicines Agency have concluded that metformin can be used with a GFR as low as 30 mL/min/m².^{60,61} Some believe, however, that there is a disproportionate fear surrounding the safety of metformin in CKD that may not be valid if patients are counseled and monitored carefully.

A retrospective comparative effectiveness study of oral antidiabetic drugs has shown that in comparison with metformin, treatment with sulfonylureas increases the risks of a decline in eGFR, ESKD, and death.⁶² Many of the sulfonylureas should be avoided in CKD owing to the risk of hypoglycemia; others should be used with caution. Newer approaches to improving IR, such as the provision of sodium glucose cotransporter 2 (SGLT-2) inhibitors and/or incretin-based therapies, provide novel strategies for diabetic kidney disease. Because recent large randomized trials have shown that empagliflozin, an inhibitor of SGLT-2, decreases the risk for composite cardiovascular events⁶³ and slows the progression of diabetic kidney disease,⁶⁴ this new class of drugs seems promising and may be a valuable addition to the current armamentarium of strategies to prevent the progression of diabetic kidney disease. Although dipeptidyl peptidase IV inhibitors are well tolerated in general, dosage adjustments according to kidney function are needed to avoid side effects.

HYPOTHALAMOPITUITARY AXIS

THYROID HORMONAL ALTERATIONS

Although thyroid hormones are necessary for the growth and development of the kidney and for the maintenance of water and electrolyte homeostasis, the kidney is involved in the metabolism and clearance of these hormones. Thus a decline of kidney function is accompanied by a characteristic disturbance in thyroid physiology (Table 56.2).

CAUSES OF THYROID HORMONE DISTURBANCES IN CHRONIC KIDNEY DISEASE

The kidney contributes to the clearance of iodide. Plasma iodide retention in CKD favors thyroidal iodide uptake and potentially blocks thyroid hormone production by a negative feedback mechanism.^{65,66} Serum free triiodothyronine

Table 56.2 Thyroid Abnormalities in Chronic Kidney Disease

Site	Abnormality
Hypothalamus	Normal or high TSH Altered TSH circadian rhythm Altered TRH and TSH clearance
Pituitary gland	Increased thyroid volume Higher prevalence of goiter and hypothyroidism Low or normal total T ₃ and total T ₄ Low or normal free T ₃ and free T ₄ Impaired T ₃ conversion from T ₄ Normal total rT ₃ and elevated free rT ₃ Alteration in binding proteins Elevated serum iodine due to reduced renal excretion
Cell	Reduced thyroid hormone cell uptake Impaired binding of thyroid hormone receptor to DNA

rT₃, Reverse triiodothyronine level; T₃, triiodothyronine; T₄, thyroxine; TRH, TSH-releasing hormone; TSH, thyroid-stimulating hormone.

(T₃) concentrations in uremia may also be low, serving as an appropriate compensatory response aimed at reducing energy expenditure and minimizing protein catabolism in the presence of PEW.⁶⁷ Metabolic acidosis^{68,69} and systemic inflammation^{70–72} are additional features of uremia that can contribute further. Medications that are able to suppress thyroid hormone metabolism include corticosteroids, amiodarone, propranolol, and lithium.^{73,74}

Serum thyrotropin (thyroid-stimulating hormone [TSH]) concentrations are usually normal or elevated in CKD, but the response to TSH-releasing hormone (TRH) is generally diminished.⁷⁵ Both TSH circadian rhythm and TSH glycosylation are altered in CKD. The latter may compromise TSH bioactivity. Because serum TSH concentrations are frequently in the normal range, uremic patients are often considered euthyroid. Free and total thyroxine (T₄) concentrations may be normal or slightly reduced, mainly as a result of impaired hormone binding to serum carrier proteins. Circulating thyroid hormones are normally bound to thyroid hormone-binding globulin (TBG) and, to a lesser extent, to prealbumin (transthyretin) and albumin. Retained substances in CKD may inhibit hormone binding to these proteins. For example, urea, creatinine, indoles, and phenols all strongly inhibit protein binding of T₄. The transient elevation in plasma T₄ levels that occurs during the HD procedure may be due to the effect of the heparin used to prevent clotting in the hollow-fiber dialyzer and associated tubing, because heparin inhibits binding of T₄ to its binding proteins.⁷⁶ Most patients with ESKD have decreased plasma levels of free T₃ (low T₃ syndrome), which primarily reflects diminished conversion of T₄ to T₃ in the periphery.⁷⁷ This peculiar hormonal profile, however, is not associated with the increased conversion of T₄ to the metabolically inactive reverse T₃ (rT₃) because plasma rT₃ levels are typically normal in uremia. Such a finding differentiates the uremic patient from patients with other chronic illnesses.

In addition, the bioavailability and cell uptake of thyroid hormones may be partially blunted in uremia, leading to a state of thyroid resistance. This possibility is important to take into consideration because the serum TSH level may not be an accurate measure of the cellular action of thyroid hormone. In normal rat hepatocytes, treatment with serum from uremic patients reduced T_4 uptake by 30%.⁷⁸ Uremic plasma from patients undergoing HD inhibited the binding of thyroid hormone receptor to DNA and impaired T_3 -dependent transcriptional activation.⁷⁹ Because dialysis per se corrected these abnormalities, the investigators suggested that a dialyzable substance was involved.⁷⁹

As a consequence, CKD is associated with a higher prevalence of primary hypothyroidism, often subclinical.^{80,81} Among incident dialysis patients, 18% of peritoneal dialysis and 22% of hemodialysis patients had hypothyroidism. The prevalence of the low T_3 syndrome (reduced T_3 in the presence of normal levels of TSH and T_4) is, in comparison, remarkably high, being reported in more than 70% of patients with ESKD.^{82,83}

CLINICAL IMPLICATIONS AND CONSEQUENCES OF THYROID HORMONE ALTERATIONS IN CHRONIC KIDNEY DISEASE

Because kidney function in hypothyroid patients was reported to be maintained or improved after thyroid hormone supplementation,^{84–86} it has been speculated that thyroid disorders may impair kidney function and vice versa. Two observational studies with longitudinal design in patients with subclinical hypothyroidism and preexisting CKD have shown that T_4 supplementation attenuated the decline in kidney function over time.^{87,88} The previously mentioned studies may be limited by confounding by indication, but their findings justify the need to explore this interesting possibility.

Hypothyroidism-associated outcomes has recently been the subject of investigation in various studies of dialysis patients. These studies reported an increased risk of mortality^{89–92} and impaired health-related quality of life across energy, fatigue, physical function, and pain domains.⁹³

Subclinical hypothyroidism, or the low T_3 syndrome, on the other hand, may constitute an intermediate link between the inflammatory stress, subsequent PEW, and impaired cardiovascular response in CKD. A number of studies have consistently shown that low T_3 concentrations are inversely correlated with markers of systemic inflammatory response and are an independent predictor of mortality in euthyroid patients with ESKD (Fig. 56.2)⁹⁴ and in dialysis populations,^{95–97} having a stronger association with cardiovascular death.^{98,99} A low T_3 level has also been linked to impaired cardiac function and geometry,^{100,101} coronary artery calcification,^{102,103} increased intima-media thickness,¹⁰³ flow-mediated vasodilation (FMD),¹⁰⁴ and measures of systemic arterial stiffness.¹⁰⁵ Low T_3 levels before kidney transplantation are associated with decreased graft survival.¹⁰⁶ The observational nature of the studies reporting these results emphasizes the importance of verifying whether uremic patients without primary thyroid dysfunction would benefit from thyroid hormone therapy. So far, there is insufficient evidence to recommend routine provision of thyroid hormone replacement in CKD with low T_3 alone. Because of potential unfavorable effects of thyrotoxicosis, such as tachycardia and loss of skeletal muscle and bone, the key therapeutic approach for the successful management of nonthyroidal illness in CKD might be simply to

restore thyroid hormone deficiencies and maintain thyroid hormones within the normal range.

An early interventional study has shown that the intake of physiologic doses of T_3 (50 mcg/day) decreases thyrotropin levels and results in a borderline negative nitrogen balance (increased protein catabolism) in patients with ESKD.¹⁰⁷ This may be the natural consequence of restoring thyroid function and, in our opinion, may be easily counteracted by increasing protein intake. Before trials are conducted, other indirect approaches that may serve as proofs of concept include correcting acidosis,¹⁰⁸ oxidative stress,¹⁰⁹ or selenium deficiency.¹¹⁰ In a placebo-controlled study of 30 euthyroid patients undergoing HD, exogenous T_4 administration over 3 months reduces lipoprotein(a) and total and low-density-lipoprotein cholesterol levels, without evidence of thyrotoxicosis.¹¹¹ However, in a large observational study of patients with ESKD, hypothyroid patients receiving exogenous thyroid hormones were at the same risk for death compared with those without medication.¹¹²

GROWTH HORMONE

The growth hormone (GH)/insulin-like growth factor 1 (IGF-1) system is of key importance for anabolism, body growth, and body composition. It regulates a range of metabolic processes that are needed for the growth of cells and tissues in the body during all phases of life, but with the most profound effects during childhood. The metabolism and secretion of GH, a 22-kDa, 191-amino acid protein produced in the pituitary gland, are inhibited by somatostatin and stimulated by GH-releasing hormone, but many other factors are also involved, such as fatty acids and other nutrients and factors linked to nutrient intake, such as ghrelin, leptin, and neuropeptide Y. In general, nutrient intake regulates GH secretion so that body protein stores rather than fat tissue are preserved, especially during energy restriction. Fasting, as well as insulin-induced hypoglycemia, increases GH secretion, whereas a glucose load decreases circulating GH by reducing somatostatin release. On the other hand, the supply of protein and amino acids, especially arginine, increases GH secretion. During CKD, many of these pathways may be disturbed.^{113–115}

GH is an anabolic hormone stimulating protein synthesis, bone growth, calcium retention, bone mineralization, and lipolysis, with decreases in body fat (Fig. 56.3). Although the effects of GH may vary according to whether the patient is fasting or fed, GH reduces hepatic glucose uptake and promotes gluconeogenesis and lipolysis, thereby opposing the glucose-lowering actions of insulin. GH released from the pituitary acts in an endocrine fashion on hepatic GH receptors to trigger the synthesis and release of IGF-1 from the liver. IGF-1 circulates free (biologically active) or bound to proteins (IGF-binding proteins [IGFBPs] 1–6). The binding of IGF-1 to specific muscle receptors induces muscle synthesis, inhibits muscle proteolysis, promotes the delivery of amino acids and glucose to myocytes, and stimulates myoblast proliferation.^{113–115} Disturbances in the GH/IGF-1 system may therefore contribute to many complications in CKD, such as growth retardation, PEW and sarcopenia, and progression (i.e., loss of kidney function in CKD). GH deficiency is associated with decreases in GFR and renal plasma flow. Low circulating levels of IGF-1 are associated with increased

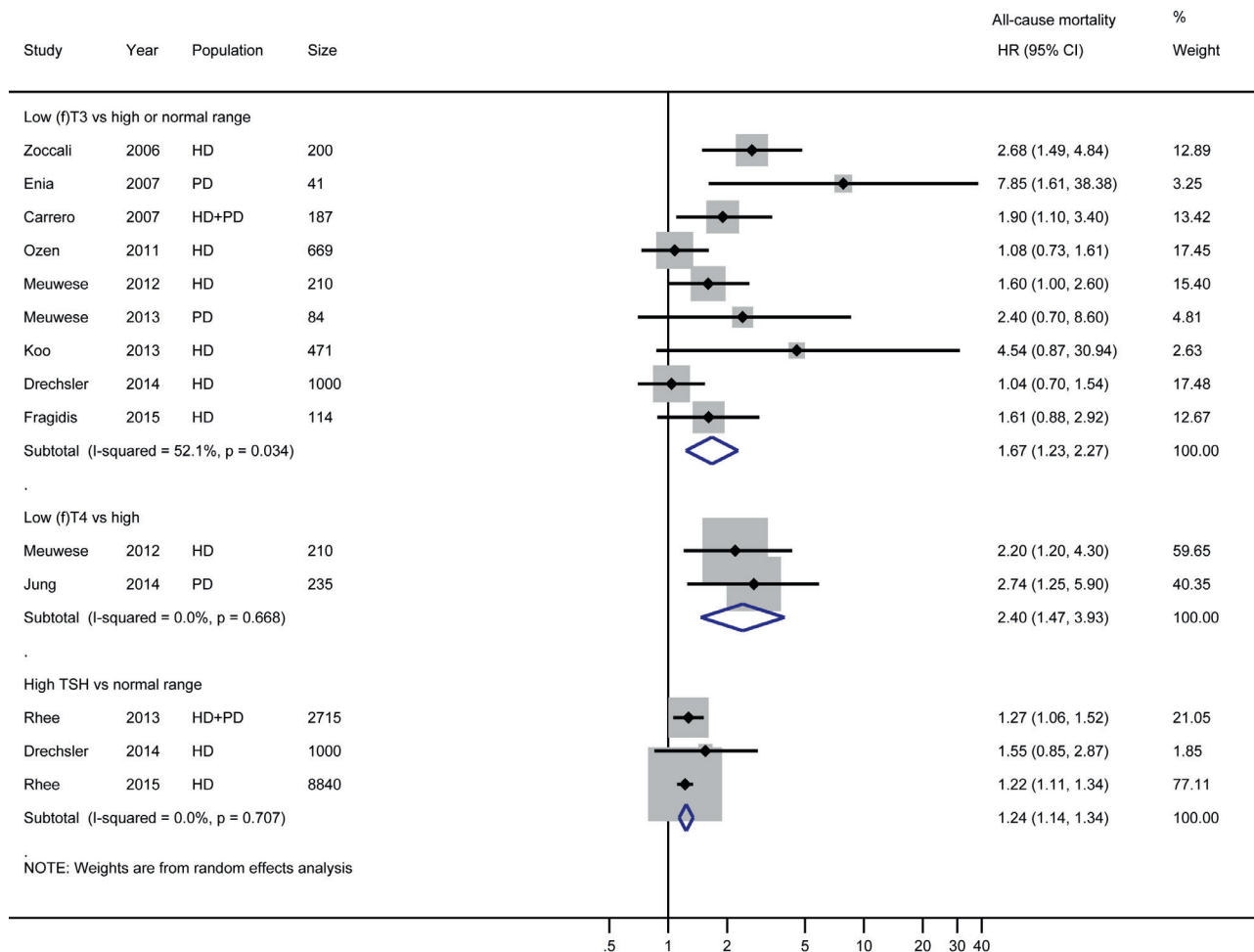


Fig. 56.2. Thyroid alterations and mortality. This Forest plot depicts the association between various forms of thyroid function test result derangements and risk for all-cause mortality, with a meta-analysis using the DerSimonian and Laird random-effects model. All hazard ratios (HRs) are based on the most fully adjusted reported model. *CI*, Confidence interval; *(f)T3*, free triiodothyronine; *(f)T4*, free thyroxine; *HD*, hemodialysis; *PD*, peritoneal dialysis; *TSH*, thyrotropin. (From Xu H, Brusselsaers N, Lindholm B, et al. Thyroid function test derangements and mortality in dialysis patients: a systematic review and meta-analysis. *Am J Kidney Dis*. 2016;68[6]:923-932.)

mortality in patients with CKD stage 5 at the time of dialysis initiation.¹¹⁶ Whereas the therapeutic use of GH to promote growth in growth-retarded children with advanced CKD is an accepted therapy, the administration of GH or IGF-1 might also improve nutritional status in adults with CKD. Furthermore, increased circulating levels of GH and IGF-1 may acutely improve kidney function in children and adults.^{117–119} New therapeutic tools such as IGF-1 receptor inhibitors provide the possibility for modulating the IGF system to treat patients with kidney disease—for example, those with diabetic nephropathy.¹²⁰ However, it should be noted that experimental studies in mice have suggested that GH and IGF-1 may increase the risk of glomerular sclerosis and could thereby contribute to the progression of CKD.^{121,122}

RESISTANCE TO GROWTH HORMONE IN CHRONIC KIDNEY DISEASE

Because growth retardation is common even though serum GH concentrations are normal or even elevated in children with CKD, a state of GH resistance and possibly also IGF-1 resistance has been proposed.^{113,123,124} Insensitivity to GH is

the consequence of multiple defects in the GH/IGF-1 system including, at the molecular level, a defect in JAK/STAT phosphorylation that may be due in part to concurrent inflammation.^{125,126} One clinical implication of this resistance to GH and IGF-1 is that children with advanced CKD whose growth has been impaired often require very large doses of GH to achieve normal or near-normal body growth.

Resistance to the actions of GH and IGF-1 is also typically present in adult patients with advanced CKD.¹¹⁴ This may be due to decreased GH receptors and/or post-GH receptor defects, as well as to decreased IGF-1 synthesis. Whereas serum IGF-1 levels in general are low, the circulating concentrations of IGFBPs are often elevated.¹²⁰ IGF-1 bioavailability may be reduced, because of the following: (1) reduced synthesis of IGF-1 receptors in the muscle¹²⁷; (2) inactivation of IGF-1 due to increased binding to IGFBPs; and (3) increased hepatic production of IGFBPs (IGFBP-1 and IGFBP-2) and reduced excretion of IGFBPs in general, leading to a larger proportion of inactive IGF-1, despite normal total serum IGF-1 concentrations.^{128,129} Newer treatment modalities targeting GH resistance with recombinant human IGF-1 (rhIGF-I),

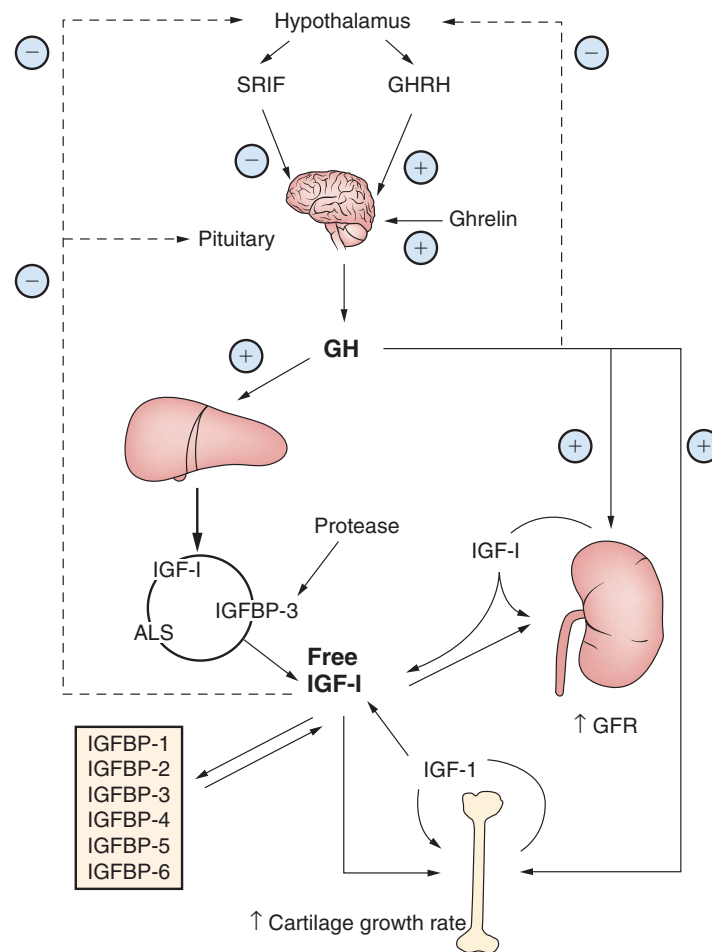


Fig. 56.3 Deranged somatotrophic axis in chronic renal failure. The growth hormone/insulin-like growth factor-1 (GH/IGF-1) axis in chronic kidney disease (CKD) is markedly different from the normal axis. In CKD, the total concentrations of the hormones in the GH/IGF-1 axis are not reduced, but there is reduced effectiveness of endogenous GH and IGF-1, which probably plays a major role in reducing linear bone growth. The reduced effectiveness of endogenous IGF-1 likely is due to decreased levels of free, bioactive IGF-1 as levels of circulating inhibitory IGF binding proteins (IGFBPs) are increased. ALS, Acid-labile subunit protein; GFR, glomerular filtration rate; GHRH, growth hormone-releasing hormone; SRIF, somatotropin release-inhibiting factor. (Modified from Roelfsema V, Clark RG: The growth hormone and insulin-like growth factor axis: its manipulation for the benefit of growth disorders in renal failure. *J Am Soc Nephrol.* 2001;12:1297–1306.)

recombinant human IGFBP3 (rhIGFBP3), and IGFBP displacers may prove to be more effective in treating growth failure in CKD.¹¹⁴ Finally, it has been elegantly shown that resistance to pharmacologic doses of GH may be related not to uremia per se, but rather to an increased inflammatory state associated with uremia.¹³⁰ Abnormalities in the interaction of these pathways with those that involve other molecules, such as ghrelin, myostatin, and the SOCS family, may also be important.¹¹³

GROWTH FAILURE IN CHILDREN WITH CHRONIC KIDNEY DISEASE

Recombinant human GH (rhGH) is an approved treatment for growth failure in children with kidney failure that has proven to be safe and efficacious. Identifying and addressing growth failure early on is an important component in the treatment of children with CKD. Treatment with rhGH is used in approximately 15% of all children undergoing dialysis in the United States.¹³¹ Unfortunately, many children with CKD and growth retardation still do not receive adequate

GH treatment for their growth failure. GH therapy should be considered in children with CKD who have a height less than 2 standard deviations (SDs) below the mean. Unusual causes for poor growth, such as hypothyroidism, should be investigated. Early institution of GH therapy is likely to improve the final achieved height. rhGH is administered as a daily subcutaneous injection. Once treatment is initiated, monitoring of growth, pubertal stage, nutritional state, fundoscopic examination (to detect papilledema due to intracranial hypertension), and blood examination should occur every 3 to 4 months to determine whether growth is adequate and whether dose adjustments are needed. In patients younger than 3 years, the head circumference should be routinely monitored as well. Later studies have shown that rhGH treatment is most effective when started at an early age, and that the growth response is affected by the degree of impairment of kidney function.¹³²

Even though rhGH has been shown to improve “catch-up” growth, the final adult height may still be below the genetic target. After kidney transplantation, growth retardation may

persist because of multiple factors, such as corticosteroid use, decreased kidney function, and an abnormal GH/IGF-1 axis.¹¹⁴ Although there have been concerns that long-term therapy with rhGH may have various adverse effects, rhGH is generally very well tolerated and does not seem to be associated with an increased incidence of glucose intolerance, pancreatitis, progressive deterioration of kidney function, acute allograft rejection, or fluid retention.^{133–135} Newer formulations of rhGH have been undergoing experimental testing with the hope that adverse effects could be further reduced and efficacy increased, and the administration schedule might be more convenient in comparison with currently available formulations.

GROWTH HORMONE TREATMENT IN ADULT PATIENTS WITH CHRONIC KIDNEY DISEASE

Many studies have explored a possible therapeutic role of rhGH and rhIGF-1 therapy in the CKD population. Among patients undergoing dialysis, evidence has suggested that rhGH stimulates protein synthesis, decreases urea generation, and improves nitrogen balance,¹²³ effects that appear to be dose-dependent.¹³⁶ IGF-1 enhances the intracellular transport of glucose and amino acids, stimulates protein synthesis, suppresses protein degradation, and stimulates bone growth and enlargement of many organs.^{115,137} Whereas the use of rhGH or rhGH plus rhIGF-1 in patients undergoing dialysis has been generally well tolerated,^{136–138} and long-term GH replacement may even improve cardiovascular mortality and morbidity in GH-deficient adults,^{139,140} rhGH treatment in patients with an acute critical illness may result in increased mortality.¹⁴¹ Among reported adverse reactions to GH treatment are a higher risk of benign intracranial hypertension, hyperglycemia, and fluid retention. In obese adults, rhGH therapy leads to a decrease in visceral adiposity and increase in lean body mass, as well as beneficial changes in the lipid profile, without inducing weight loss, despite increases in fasting plasma glucose and insulin levels.¹⁴²

GROWTH HORMONE/INSULIN-LIKE GROWTH FACTOR-1 SYSTEM AND KIDNEY FUNCTION

Disordered regulation of the IGF system is found in a number of kidney diseases, including polycystic kidney disease (PKD); IGF-1 stimulates the proliferation of cyst-lining cells from patients with PKD, an effect inhibited by rosiglitazone.¹²⁰ Receptors for GH and IGF-1 are expressed in the kidney and influence kidney structure and function,¹¹⁷ and short-term rhGH treatment in CKD has been linked to a general improvement of capillary blood flow.¹⁴³

Regarding kidney function, GH may increase renal hemodynamics and filtration rate, whereas rhIGF-1 can enhance GFR and renal plasma flow when administered short term to humans with ESKD.¹¹⁸ In connection with this, the GFR and renal plasma flow rates are elevated in patients with acromegaly, whereas kidney function is usually low in GH-deficient states. GH increases the GFR with a delay of many hours, up to 1 day, consistent with the induction of IGF-1 synthesis. Endogenous IGF-1 may contribute to the physiologic regulation of GFR.¹⁴⁴

Several studies have assessed the potential of rhGH and/or rhIGF-1 administration to improve kidney function. Whereas GH resulted in no or only a modest and transitory increase of the GFR in adults with advanced CKD and in

children with growth failure,¹¹⁷ IGF-1 produced a more sustained increase in GFR and renal plasma flow.¹¹⁸ A regimen using rhIGF-1 in patients with advanced CKD was well tolerated and resulted in a sustained improvement in kidney function.¹¹⁹ Transgenic mice expressing GH developed increased mesangial proliferation followed by progressive mesangial sclerosis, which was not seen in transgenic mice expressing IGF-1,^{121,122} and there has been a concern that GH/IGF-1 therapy could contribute to the progression of CKD. However, prolonged treatment with GH in children with CKD has not been reported to lead to more rapid progression of CKD.^{133,134}

PROLACTIN

Prolactin's normal function in women is to promote lactation, but its function in men has not been fully established. Serum prolactin concentrations are usually elevated in patients with CKD, and the prevalence of hyperprolactinemia in ESKD ranges between 30% and 65%.^{145–147} Hyperprolactinemia in CKD is understood as a consequence of reduced renal clearance¹⁴⁷ and increased production due to suppressed dopaminergic activity.¹⁴⁸ Thus antidopaminergic medications (e.g., neuroleptics, metoclopramide, cimetidine), which can further stimulate prolactin production, should be minimized or avoided if possible. The consequences of hyperprolactinemia in CKD and ESKD are reflected in the commonly observed reproductive abnormalities resulting from the associated inhibition of gonadotropin secretion. Hyperprolactinemic patients eventually experience galactorrhea and infertility due to the inhibition of gonadotropin secretion; in women, amenorrhea may concur, whereas in men, erectile dysfunction and hypogonadism often appear concomitantly. Although bromocriptine treatment has been proven to decrease prolactin levels in uremic men and women,¹⁴⁹ the previously mentioned symptoms do not fully disappear, suggesting that other factors may contribute in parallel. Therapy with ESAs has been suggested to decrease serum prolactin levels¹⁵⁰ and improve sexual function.¹⁵¹ Thus it has been postulated that prolactin may contribute to the severity of anemia associated with CKD.

It is possible that prolactinemia may have previously unrecognized effects independent of its effects on the gonads. Studies in patients with CKD have reported a strong association among prolactinemia, endothelial dysfunction, arterial stiffness, and cardiovascular outcomes in men and in women.¹⁵² These associations may be explained as a consequence of the following: (1) the inhibition of gonadotropic hormones, which may be linked per se to increased cardiovascular risk (discussed later); (2) decreased dopaminergic activity; (3) other risk factors affecting prolactin production, such as hypercytokinemia¹⁵³; or (4) yet unknown mechanisms. Advancement in the understanding of prolactin physiology reveals additional functions, such as regulating the immune system and serving as a growth and antiapoptotic factor. As a growth factor, prolactin influences hematopoiesis, angiogenesis, and blood clotting. Prolactin modulates the inflammatory response, stimulates the adhesion of mononuclear cells to endothelium, and enhances vascular smooth muscle cell proliferation.^{154–157} Prolactin also induces regional vasoconstriction through β_2 -adrenergic and nitric oxide mechanisms.¹⁵⁸ Some small studies have evaluated the

effects of bromocriptine therapy in CKD, describing a reduction in blood pressure and the regression of left ventricular hypertrophy in patients undergoing dialysis.^{159–161} Another randomized-controlled trial in patients with diabetes mellitus and stage 4 CKD tested bromocriptine supplementation for 6 months versus placebo¹⁶²; decreases in blood pressure and left ventricular mass index were observed. Whether these effects were, at least partly, mediated by prolactin reduction is unknown.

ADRENAL GLANDS

Because symptoms of hypercortisolism and hyperaldosteronemia are common in CKD, it has been proposed that the hypothalamic-pituitary-adrenal (HPA) axis may be upregulated.¹⁶³ This proposal is plausible, given that both glucocorticoid and aldosterone metabolites are excreted by the kidneys and that cortisol metabolism is partly regulated by the kidneys. Nonetheless, few studies to date have addressed adrenal gland disorders in CKD, an issue in part hampered by the prescription of medication interfering with the RAAS and cortisol system.

ADRENOCORTICOTROPIC HORMONE

Adrenocorticotrophic hormone (ACTH) was used 60 years ago for the treatment of nephrotic syndrome in children but was gradually replaced by synthetic glucocorticoid analogs. In addition to its role in controlling steroidogenesis, ACTH stands as a physiologic agonist of the melanocortin system. Clinical and experimental evidence suggests that ACTH may have antiproteinuric, lipid-lowering, and renoprotective properties, which are not fully explained by its steroidogenic effects.^{164,165}

ALDOSTERONE AND CORTISOL

Over and above the classical effect on sodium reabsorption, aldosterone may exert other effects on renal and cardiovascular damage. Aldosterone increases oxidative stress and promotes vascular inflammation^{166,167} and impairs vascular reactivity by limiting the bioavailability of nitric oxide.¹⁶⁸ In the presence of salt overload, aldosterone causes hypertrophy and fibrosis in the heart, both of which are prevented by the administration of mineralocorticoid receptor (MR) antagonists.¹⁶⁹ The MR binds aldosterone and cortisol with similar affinities. Under normal conditions cortisol is incapable of activating the MR because cortisol is converted into the inactive metabolite cortisone by 11 β -hydroxysteroid dehydrogenase type 2 (11 β -HSD2).¹⁷⁰ Thus it may be hypothesized that the beneficial effects of MR antagonists may result from blocking the action of both aldosterone and cortisol. In an observational analysis in patients with type 2 diabetes mellitus undergoing maintenance HD,¹⁷¹ the joint presence of high serum aldosterone and high serum cortisol concentrations was associated with sudden cardiac death. Whether the use of MR antagonists decreases the risk of sudden death in such patients must be examined in future trials.

ADRENAL ANDROGENS

Dehydroepiandrosterone (DHEA) and dehydroepiandrosterone sulfate (DHEA-S) are secreted from the zona reticularis of the adrenal gland. DHEA and DHEA-S are interconverted, and DHEA serves as a precursor of sex hormones. In a

population-based study of young adults, serum DHEA concentrations were inversely associated with kidney function.¹⁷² Low circulating serum concentrations of DHEA-S were also associated with the progression of glomerular injury in men with type 2 diabetes mellitus.¹⁷³ This finding is in line with those of two studies showing that serum DHEA-S was significantly reduced in male patients on maintenance HD and was associated with all-cause and CVD-related mortality.^{174,175} Like concentrations of other hormones, the reduced DHEA-S concentration may be a surrogate of disease severity, being suppressed in critical illness. However, DHEA-S may have protective functions against atherosclerosis and CVD, given that DHEA supplementation improves endothelial function and insulin sensitivity in men.¹⁷⁶ DHEA-S may have other functions as a peroxisome proliferator-activated receptor α (PPAR α) activator that can modulate immune function, inflammation, and oxidative stress.¹⁷⁷ Finally, one should not forget that DHEA-S may be an intermediate in the pathways of both prolactin (upstream) and testosterone (downstream), thus mediating the risk.

GONADAL DYSFUNCTION

Disturbances in the hypothalamic-pituitary-gonadal axis are common in patients with CKD and play an important role in the development of sexual dysfunction (Table 56.3). Sexual dysfunction in these patients should be thought of as a multifactorial problem that is affected by a variety of physiologic and psychological factors, as well as comorbid conditions. In addition to a number of endocrine alterations described later, diabetes and vascular disease, for example, can interfere with the ability of the male patient to achieve an erection and the female patient to achieve sexual arousal. Various psychological factors, such as depression, can significantly and adversely affect sexual function in both men and women.

IN WOMEN

Endocrine Abnormalities

Elevated serum concentrations of prolactin (see earlier discussion), follicle-stimulating hormone (FSH), and luteinizing hormone (LH) are usual findings in uremic women.

Table 56.3 Endocrine Abnormalities Leading to Sexual Dysfunction in Chronic Kidney Disease

Gender Affected	Endocrine Abnormality
Men	Decreased production of testosterone Blunted increase in serum luteinizing hormone (LH) Decreased amplitude of LH secretory burst Variable increase in follicle-stimulating hormone (FSH) Increased prolactin
Women	Anovulatory menstrual cycles Lack of midcycle surge in LH Increased prolactin

Disturbances in menstruation and fertility are commonly encountered, usually leading to amenorrhea by the time the patient is diagnosed with ESKD. The menstrual cycle typically remains irregular after the initiation of dialysis. Ovarian dysfunction in women undergoing dialysis is characterized by the absence of cyclic gonadotropin and estradiol release, which result in the lack of progestational changes in the endometrium.¹⁷⁸ A midcycle LH surge cannot be mitigated with the endogenous administration of estrogen, confirming a central hypothalamic derangement.¹⁷⁹ As a consequence, anovulation and subsequent infertility are probably the major menstrual abnormalities in uremic women, together with decreased libido and reduced ability to reach orgasm.¹⁸⁰ Therefore, successful conception with pregnancy is rare in women with ESKD. Pathologic endometrium morphology is very common in uremic women of reproductive age undergoing HD, with proliferative changes in 30% and atrophic changes in almost 25%. However, it seems that the endometrium has preserved normal reactivity to circulating estrogens.¹⁸¹

Clinical Manifestations

Young uremic women usually experience premature menopause, approximately 4.5 years earlier on average than their healthy counterparts. Hypogonadism in women has been linked with sleep disorders, depression, urinary incontinence and, in the long term, osteoporosis, impaired cognitive function, and increased cardiovascular risk.¹⁸² Up to 65% of women undergoing dialysis report problems with sexual function, and up to 40% no longer engage in sexual intercourse.¹⁸³ Loss of libido may also contribute to infertility.

Finally, megestrol acetate has successfully been used in patients with ESKD as an effective therapy to treat PEW.^{184–187} This finding, together with the observation of an attenuation of the symptoms associated with uremic anorexia in women in comparison with men,¹⁸⁸ suggests the existence of a yet uncharacterized pleiotropic role for sex hormones in the regulation of nutrient homeostasis in uremia.

Treatment

General principles of treatment include education about sexual function in the setting of CKD, adequate dialysis delivery, and treatment of underlying depression. Changes in lifestyle, such as smoking cessation, strength training, and aerobic exercises, may decrease depression, enhance body image, and have positive effects on sexuality.¹⁷⁹ Limited evidence has indicated that CKD alters the pharmacokinetics of estradiol. Free and total estradiol plasma concentrations are higher in women with ESKD after an oral estradiol dose, but no change occurs in estrone concentrations. Neither estradiol nor estrone is removed in the dialysate.¹⁸⁹ Steady-state pharmacokinetics of oral estradiol has shown that women with ESKD should receive approximately 50% of the typically prescribed dosages.¹⁹⁰ No information is available on the pharmacokinetics of any of the progestins in CKD.

Chronic anovulation and lack of progesterone secretion in uremic women may be treated with oral progesterone. Because ongoing menses can contribute to the anemia of CKD, particularly in patients with menorrhagia, administration of progesterone at the end of the menstrual cycle is preferred. At present, it is not clear whether unopposed estrogen stimulation (due to anovulatory cycles) predisposes women with CKD to endometrial hyperplasia or endometrial cancer.

Thus routine gynecologic follow-up is recommended for such patients. Low estradiol levels in amenorrheic women undergoing dialysis can lead to vaginal atrophy and dyspareunia; topical estrogen cream and vaginal lubricants may be helpful in these patients. Uremic women who are menstruating normally should be encouraged to use birth control. Estradiol hormonal replacement therapy was able to restore regular menses and improve sexual function in premenopausal, estrogen-deficient women undergoing dialysis¹⁹¹ and to improve bone histomorphometry in animal models of uremia.¹⁹² Estrogen administration may positively affect sexual desire and prevent bone mass loss in postmenopausal women with ESKD.^{193,194} Hypoactive sexual desire disorder is the most commonly reported sexual problem in women with CKD, and testosterone replacement therapy has shown effectiveness in patients without renal disease.^{195,196} Nonetheless, successful kidney transplantation is clearly the most effective means to restore normal sexual desire in women with CKD.

Female Sex Hormones and Progression of Chronic Kidney Disease

Because the progression rate of renal disease is in general faster for men than for women,¹⁹⁷ it has been suggested that this gender dimorphism may be explained by the interaction of circulating steroids with specific receptors in the kidney. In experimental animal models, endogenous estrogens have shown antifibrotic and antiapoptotic effects in the kidney,^{198,199} and exogenous estradiol in ovariectomized rats attenuated glomerulosclerosis and tubulointerstitial fibrosis²⁰⁰ by protecting podocytes against injury through the upregulation of estrogen receptor β .²⁰¹ A direct extrapolation from these animal studies would suggest that exogenous estrogen administration may slow CKD progression. However, clinical evidence in this regard has been elusive, with evidence suggesting that both estrogen replacement therapy and oral contraceptives are associated with albuminuria, increased creatinine clearance, and loss of kidney function.^{202–206} The implications of these studies need to be carefully considered in the context of their observational and, in most of the cases, retrospective nature. It should also be noted that in general, hormone replacement therapy is prescribed less frequently to postmenopausal patients with ESKD than to the general population,²⁰⁷ possibly incurring a selection and underrepresentation bias.

IN MEN

Endocrine Abnormalities

The occurrence of testosterone deficiency is estimated to vary from 6% to 9.5% in community-dwelling men aged 40 to 75 years, rising to 15% to 30% in diabetic or obese men.^{208,209} In men with CKD, this prevalence is much higher, ranging between 50% and 75%.^{210–213} This deficiency is true for both free and total serum testosterone, although the binding capacity and levels of sex hormone-binding globulin seem within normal range. Causes of hypogonadism in CKD are multiple and must be sought within hyperprolactinemia (see preceding discussion), comorbid conditions (e.g., PEW, obesity, diabetes mellitus, hypertension), and medications that may influence gonadal function (e.g., ACE inhibitors, ARBs, spironolactone, ketoconazole, glucocorticoids, statins, cinacalcet).

The plasma concentration of LH is typically elevated in men with uremia, mainly owing to changes in the pulsatile release of gonadotropin-releasing hormone and LH itself, diminished feedback inhibition of LH production (because of low testosterone levels), and impaired renal clearance. FSH secretion is also increased in men with CKD, although to a more variable degree, so that the LH/FSH ratio is typically increased. Feedback inhibition of FSH is impaired because of a decrease in the peptide inhibin, which is produced by the Sertoli cells.¹⁷⁹ All these disturbances can be detected with only moderate reductions in the GFR and progressively worsen in parallel with the progression of CKD. These disorders rarely normalize with the initiation of dialysis. Instead, they often progress. A well-functioning kidney transplant is likely to restore normal sexual activity, although some features of reproductive function may remain impaired.

Clinical Manifestations

Symptoms and signs of hypogonadism in men depend on the stage in life at which hypogonadism develops and its duration. In adults with ESKD, hypogonadism, low testosterone levels, and hyperprolactinemia are likely responsible for decreased libido, erectile dysfunction, oligospermia and infertility, osteopenia and, to some extent, osteoporosis and anemia. Erectile dysfunction has been reported in 70% to 80% of men with CKD and ESKD.²¹⁴ Additional risk factors that increase the probability of erectile dysfunction are advanced age, diabetes, hypertension, dyslipidemia, smoking, and anxiety. Semen analysis typically shows a decreased volume of ejaculate, either a low sperm count or complete azoospermia, and a low percentage of motility. Medications frequently used in treating patients with CKD, such as diuretics, antihypertensive and antidepressant agents, and histamine H₂ blockers, can contribute to erectile dysfunction. Other drugs, such as spironolactone, ketoconazole, glucocorticoids, and cimetidine, can interfere directly with the synthesis of sex hormones.²¹⁵ Autonomic nervous system dysfunction, a frequent finding in patients with CKD (especially those with diabetes mellitus), likely also contributes to sexual abnormalities in CKD.²¹⁶ Disturbances in the pelvic autonomic nervous system can decrease sensation and arousal of stimuli during sexual activity. Autonomic neuropathy can also interfere with the complex neurologic axis that is necessary for the achievement of an adequate erection.

Uremic hypogonadism may contribute to sarcopenia. In physiologic conditions, testosterone is an anabolic hormone that plays an important role in inducing skeletal muscle hypertrophy by promoting nitrogen retention, stimulating fractional muscle protein synthesis, inducing myoblast differentiation, and augmenting the efficiency of amino acid reuse by skeletal muscles.²¹⁷ Consequently, positive associations between serum free and total testosterone levels with creatinine and hand grip strength have been reported in men undergoing dialysis,^{218,219} and endogenous testosterone in men with CKD stages 3 to 4 emerged as an important determinant of both muscle mass and strength.²²⁰ Interventional studies with nandrolone decanoate (a testosterone synthetic agonist) in dialysis populations have shown significant improvements in muscle mass and strength, as well as in nutritional status.^{221,222}

Testosterone is known to exert a stimulatory effect on erythropoiesis²²³ by inducing the growth of differentiated

stem cells and enhancing the sensitivity of erythroid progenitors to circulating erythropoietin (EPO).^{224,225} Because CKD is more common in older men, it is likely that both an age-associated decline in testosterone and the endocrine CKD abnormalities that encompass male hypogonadism contribute to some extent to a decline in erythroid mass and may contribute to anemia. For this reason, testosterone deficiency has been proposed as an additional cause of anemia in ESA-naïve nondialyzed male patients with CKD and as an additional cause of resistance to ESAs in men treated with ESA undergoing HD.²²⁶ Before the clinical introduction of recombinant human EPO (rhEPO) in 1989, androgens were the main pharmacologic intervention for correcting the anemia of ESKD. Reciprocally, anemia in patients with ESKD has been associated with a reduction of libido and endothelial dysfunction,²²⁷ and rhEPO therapy has led to an increase in sexual desire and performance and an improvement in erectile function in some, but not all, patients.^{228,229} Correction of anemia, improved sense of well-being, and direct endocrine effects may play a role in this result.¹⁸⁰ In addition, studies evaluating changes in the health-related quality of life in response to rhEPO therapy have noted significant improvements in physical and social functioning, overall mental health, and satisfaction with sexual activity.²³⁰

Finally, hypogonadism in men with non-dialysis-requiring CKD has been associated with arterial stiffness, endothelial dysfunction, and risk of cardiovascular events.²³¹ In patients undergoing dialysis, hypogonadism has also been linked to increased mortality risk^{213,218,232} (Fig. 56.4), arterial stiffness,²³³ worse quality of life,²¹⁰ and heightened inflammation.²¹² The links between testosterone and cardiovascular complications may be explained by its association with risk factors such as dyslipidemia, obesity, diabetes mellitus, and metabolic syndrome, which may contribute to endothelial dysfunction and atherosclerosis per se. However, testosterone may also have direct atheroprotective effects in the cardiovascular system.¹²³ Transdermal testosterone therapy improved exercise-induced myocardial ischemia during an exercise stress test in men with stable angina,¹²⁴ and men with prostate cancer undergoing androgen deprivation therapy experienced an increase in central arterial pressure.¹²⁵ In animal models, testosterone supplementation inhibits neointimal plaque development,¹²⁶ stimulates endothelial progenitor cells,¹²⁷ increases nitric oxide release from vascular endothelial cells,¹²⁸ and enhances myocardial perfusion.^{129,130}

The gonadal axis is suppressed by inflammatory cytokines,^{234,235} and therefore any inflammatory disease may induce testosterone deficiency. Thus low testosterone levels could be considered a biomarker of chronic inflammatory disease. In support of this contention, studies have depicted a strong inverse association between endogenous testosterone and surrogates of inflammation in various CKD populations.^{212,218,232,233} However, it is also possible that testosterone has immunomodulatory actions per se, as suggested by the suppression of cytokine production in hypogonadal men with diabetes mellitus, coronary heart disease, and/or metabolic syndrome after supplementation with testosterone.^{236–238}

Treatment

General principles of treatment of hypogonadism in men include optimal delivery of dialysis and adequate nutritional intake, as well as screening for depressive symptoms. A study

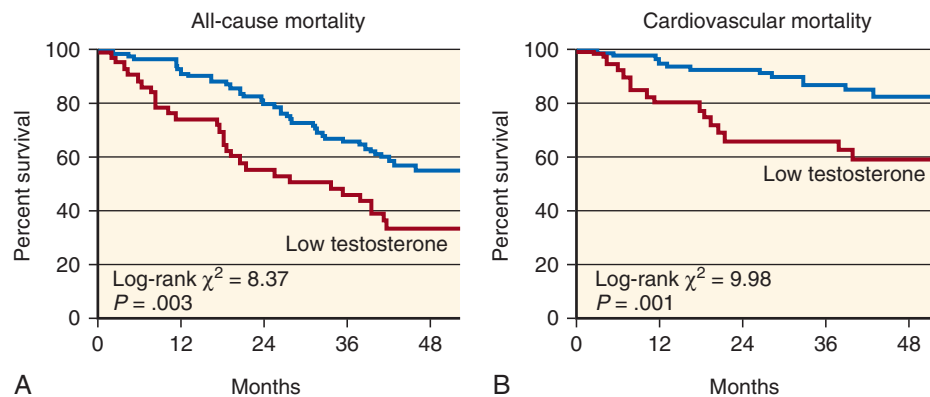


Fig. 56.4 Sex hormone alterations and outcome of chronic kidney disease (CKD). Reduced testosterone levels in male patients undergoing hemodialysis have been linked to increased mortality risk, especially that from cardiovascular causes. (Modified from Carrero JJ, Qureshi AR, Parini P, et al: Low serum testosterone increases mortality risk among male dialysis patients. *J Am Soc Nephrol*. 2009;20:613–620.)

has compared the pharmacokinetics of a testosterone patch in patients with ESKD and in hypogonadal men with normal kidney function.²³⁹ The researchers found that the half-life of testosterone after withdrawal of the patch was not different between the groups, nor was the minimum or maximum serum testosterone concentration during the period of patch application. Thus it is likely that usual doses of testosterone replacement can be used in men with ESKD, with the usual monitoring of serum testosterone concentrations, with dose adjustments as needed. Data on the use of testosterone in ESKD are limited; some studies have suggested that erectile function in ESKD does not improve with testosterone supplementation,²⁴⁰ and others have indicated that normalization of endogenous testosterone by topical gels in hypogonadal men with advanced CKD improve sexual function.²⁴¹ The daily administration of 100 mg of 1% testosterone gel for 6 months in 40 hypogonadal men with ESKD neither increased serum testosterone concentrations nor had an effect on ESA requirements.²⁴² It is possible that higher testosterone dosages may be required to achieve a clinical benefit in this patient group. Alternative modes of administration, such as intramuscular injection, may ease compliance and bioavailability. The response of testosterone in this setting may be modulated by the patient's nutritional status, activity level, and GH (and GH-binding protein) status. Because psychosocial factors may also take part in the pathophysiology of erectile dysfunction, nocturnal penile tumescence testing may be used to differentiate organic and psychological causes of impotence.

Whether testosterone deficiency may constitute a new pathophysiologic pathway in CKD-related complications such as anemia, PEW, and cardiovascular risk deserves further attention. Androgen therapy in supraphysiologic dosages has been linked to adverse effects.^{28,160} Isolated cases of increase in blood pressure and clinically significant edema in healthy older men have been reported,^{28,151,152} as has fluid retention in men undergoing HD.⁷⁵ Treated patients must be closely monitored but, again, restoration of deficiencies and maintenance of total and free testosterone concentrations within the normal range may be the key to successful treatment.

Male Sex Hormones and Progression of Chronic Kidney Disease

In line with the finding that CKD progression is faster in men than in women, animal models of kidney injury have

shown that orchidectomy attenuates glomerular and tubular damage, kidney fibrosis, and proteinuria.^{243,244} Furthermore, animal studies have described proinflammatory, proapoptotic, and profibrotic effects of supraphysiologic dosages of testosterone during acute and chronic kidney injury.^{199,245,246} On the other hand, it is difficult to reconcile such experimental findings with clinical observations, given that testosterone production decreases with progressive CKD, and that both the incidence and progression of CKD occur mainly in older men, who have lower testosterone concentrations on average. In line with this thinking, a population-based study of men has reported that impaired kidney function and low serum testosterone concentrations are additive (and independent) mortality risk factors.²⁴⁷ Thus evidence has accumulated suggesting that testosterone may be protective to the kidney.

A large case-control study involving more than 10,000 men newly diagnosed with nonmetastatic prostate cancer has demonstrated that androgen deprivation therapy increases the risk of acute kidney injury.²⁴⁸ Potential protective mechanisms include inducing vasodilation in renal vessels and enhancing the production of nitric oxide.²⁴⁹ Thus the use of androgen deprivation therapy might antagonize testosterone, raising the risk of damage to the glomerulus. An additional explanation for the role of hypogonadism on kidney injury has been demonstrated in male rats subjected to renal ischemia followed by reperfusion.²⁵⁰ In this setting, testosterone concentrations decreased dramatically after only 3 hours. Infusion of testosterone 3 hours after reperfusion attenuated the 24-hour increase in plasma creatinine and urinary kidney injury molecule-1 (KIM-1), prevented the reduction in outer medullary blood flow, and attenuated the 48-hour increase in intrarenal inflammation. Castration caused greater rises in plasma creatinine levels and KIM-1, and treatment with anastrozole (an aromatase inhibitor) plus testosterone almost normalized these markers.²⁵⁰

VITAMIN D, PARATHYROID HORMONE, AND KIDNEY DISEASE

The endocrine system of vitamin D and PTH is crucial for the homeostasis of calcium and phosphorus and for maintaining a healthy bone status. Altered levels of serum phosphate, calcium, vitamin D, PTH, and fibroblast growth factor 23

(FGF23) contribute to the complex multiorgan syndrome of CKD—mineral and bone disorder (CKD-MBD), which leads to osteoporosis, fractures, CVD, and other complications.^{251–253} Whereas abnormalities in PTH and the consequences of hyperparathyroidism are well recognized clinical problems in the management of CKD-MBD,²⁵⁴ the awareness of the importance of vitamin D deficiency has also increased.^{251–253} Indeed, the expression of vitamin D receptors (VDRs) by many different tissues and cells suggests a multifaceted role for this system.^{255,256} Physiology and management of these hormonal alterations are discussed in more detail in Chapter 53.

VITAMIN D AND PARATHYROID HORMONE: METABOLISM AND ACTIONS IN CHRONIC KIDNEY DISEASE

The vitamin D family includes the biologically inert fat-soluble prohormones—vitamin D₂ (ergocalciferol) and vitamin D₃ (cholecalciferol). These prohormones are obtained mainly from sun exposure—by photochemical conversion of 7-dehydrocholesterol in the skin, food (especially fatty fish), and nutritional supplements. These prohormones are converted by hydroxylation in the liver to 25-hydroxyvitamin D and by a second hydroxylation step (performed by the 1 α -hydroxylase) to its active form, 1,25-dihydroxyvitamin D (calcitriol). The kidney is the most abundant source (but not exclusive) of 1 α -hydroxylase in the body, and therefore CKD coincides with natural deficiencies in calcitriol.^{252,253}

In normal conditions, the renal production of calcitriol is tightly regulated by PTH and by serum calcium and phosphorus levels. FGF23, which is secreted from the bone, suppresses calcitriol synthesis, as does metabolic acidosis. Extrarenal calcitriol is converted in other tissues, including the skin, colon, prostate, and macrophages, but the regulatory processes for this conversion are not well understood. Calcitriol has a half-life of only 4 to 6 hours, and the circulating levels are low. Calcitriol, which is transported to various target organs by the vitamin D-binding protein (VDBP), mobilizes calcium into the bloodstream by promoting the absorption of calcium and phosphorus from food in the intestines and reabsorption of calcium in the kidneys. Calcitriol also promotes bone mineralization, bone growth, and bone remodeling by osteoblasts and osteoclasts and prevents hypocalcemia. Vitamin D deficiency has been linked to multiple disorders, including growth retardation, skeletal abnormalities (e.g., osteopenia, osteoporosis, increased risk of fractures), muscle weakness, left ventricular hypertrophy, and increased susceptibility to cancer, diabetes, and autoimmune and infectious diseases.^{254,257–259} The effects of vitamin D goes beyond those linked to calcium and bone homeostasis. In patients with CKD who display a combination of bone metabolism disorder, muscle wasting, and muscle weakness, vitamin D is thought to contribute to maintain musculoskeletal health.²⁶⁰ In the kidney, vitamin D may be important for maintaining podocyte function, preventing epithelial to mesenchymal transformation, and suppressing renin gene expression and inflammation.²⁶¹ Vitamin D also appears to be protective in models of diabetic nephropathy through targeting of the RAAS and the nuclear factor-kappa B (NF- κ B) pathway.²⁶²

PTH has a key role in calcium and phosphate homeostasis. It stimulates excretion of phosphate by the renal tubular

cells and indirectly modulates intestinal calcium absorption by stimulating the activation of calcitriol synthesis. As vitamin D deficiency progresses, the parathyroid glands are maximally stimulated, causing secondary hyperparathyroidism. Thus levels of 25-hydroxyvitamin D are inversely associated with PTH levels, and calcitriol inhibits PTH expression. PTH increases the metabolism of 25-hydroxyvitamin D to calcitriol, further exacerbating the vitamin D deficiency. Hyperparathyroidism has been implicated in left ventricular hypertrophy, as well as in the metabolic syndrome contributing to impaired glucose tolerance and dyslipidemia.²⁶³ PTH and serum phosphate are well-established biomarkers of CKD-MBD, and FGF23 has been gaining increased attention.²⁶⁴ However, the value of PTH²⁶⁵ and FGF23 in clinical monitoring,²⁶⁶ among others, has been questioned owing to their wide (and assay-dependent) biologic variability.

VITAMIN D DEFICIENCY IN CHRONIC KIDNEY DISEASE

Insufficient to deficient levels of 25-hydroxyvitamin D have been reported in most individuals with CKD, including patients undergoing maintenance dialysis and others with non-dialysis-requiring CKD.^{267,268} Vitamin D deficiency seems to be more pronounced in blacks.²⁶⁹ The circulating levels of 25-hydroxyvitamin D and calcitriol decrease with diminishing kidney function.^{268,270} Because 25-hydroxyvitamin D has a longer half-life than calcitriol ($\approx$ 3 weeks), it is considered the best measure of vitamin D status.²⁷¹ On these premises, vitamin D insufficiency is commonly defined as 25-hydroxyvitamin D levels lower than 30 ng/mL, whereas deficiency starts at the threshold of less than 15 ng/mL. Although there are known seasonal, geographic, ethnic, and age-related variations, the desirable level of vitamin D in patients with CKD has not yet been defined; nevertheless, it is recommended to maintain serum concentrations of 25-hydroxyvitamin D above 30 ng/mL.²⁵⁴ Most commercial assays for 25-hydroxyvitamin D are thought to be acceptable for detecting vitamin D deficiency.²⁵² However, it should be noted that there are considerable between-laboratory and interassay variations in tests for the measurement of circulating concentrations of 25-hydroxyvitamin D and PTH.²⁷² The 1,25-dihydroxyvitamin D assay should not be used to detect vitamin D deficiency in CKD because 25-dihydroxyvitamin D levels are usually low as a consequence of reduced 1 α -hydroxylase activity in this disease, but they could also be normal, or even elevated, as a result of secondary hyperparathyroidism.

VITAMIN D SUPPLEMENTATION IN CHRONIC KIDNEY DISEASE

Providing children and adults with at least 800 IU of 25-hydroxyvitamin D₃/day or its equivalent should guarantee vitamin D sufficiency. Unless a person eats oily fish frequently, it is difficult to provide that much vitamin D₃ on a daily basis from dietary sources. Thus moderate sun exposure and/or provision of supplements are needed to fulfill the vitamin D requirement in patients with CKD.²⁶⁷ Pharmacologic doses of vitamin D can be administered in the form of nutritional supplements or active vitamin D agents. The National Kidney Foundation Kidney Disease Outcomes Quality Initiative (K/DOQI) guidelines have recommended starting with nutritional

vitamin D supplements before using activated vitamin D derivatives in patients with concurrent hyperparathyroidism associated with vitamin D insufficiency.^{267,271} Although some studies have shown that vitamin D supplementation may improve physical performance and protect against the development of hyperparathyroidism, evidence is still insufficient.²⁶⁰ Six months of therapy with calcitriol or cholecalciferol in CKD patients did not improve vascular endothelial function or inflammation.²⁷³ whereas in another study, 6 months of supplemental vitamin D analogues achieved a reduction of pulse wave velocity in patients with advanced CKD.²⁷⁴ Although epidemiologic studies have generally indicated a survival benefit associated with the provision of activated vitamin D derivatives in all CKD stages^{267,275–278} and a 27% lower associated mortality risk has been found in patients receiving calcitriol or analogues,²⁷⁹ the studies are subject to confounding by indication, and recommendations regarding nutritional vitamin D supplements are thus largely opinion-based. However, existing randomized-controlled trials do not provide sufficient or precise evidence that vitamin D supplementation lowers the mortality of patients with CKD.^{267,280}

 Complete reference list available at [ExpertConsult.com](https://www.expertconsult.com).

KEY REFERENCES

7. DeFronzo RA, Alvestrand A, Smith D, et al. Insulin resistance in uremia. *J Clin Invest*. 1981;67:563–568.
23. Beddhu S, Kimmel PL, Ramkumar N, et al. Associations of metabolic syndrome with inflammation in CKD: results from the Third National Health and Nutrition Examination Survey (NHANES III). *Am J Kidney Dis*. 2005;46:577–586.
28. Raj DS, Dominic EA, Pai A, et al. Skeletal muscle, cytokines, and oxidative stress in end-stage renal disease. *Kidney Int*. 2005;68:2338–2344.
39. Chen J, Muntner P, Hamm LL, et al. The metabolic syndrome and chronic kidney disease in U.S. adults. *Ann Intern Med*. 2004;140:167–174.
42. Shinohara K, Shoji T, Emoto M, et al. Insulin resistance as an independent predictor of cardiovascular mortality in patients with end-stage renal disease. *J Am Soc Nephrol*. 2002;13:1894–1900.
49. Pupim LB, Flakoll PJ, Majchrzak KM, et al. Increased muscle protein breakdown in chronic hemodialysis patients with type 2 diabetes mellitus. *Kidney Int*. 2005;68:1857–1865.
50. Mustata S, Chan C, Lai V, et al. Impact of an exercise program on arterial stiffness and insulin resistance in hemodialysis patients. *J Am Soc Nephrol*. 2004;15:2713–2718.
56. Ramirez SP, Albert JM, Blayney MJ, et al. Rosiglitazone is associated with mortality in chronic hemodialysis patients. *J Am Soc Nephrol*. 2009;20:1094–1101.
62. Hung AM, Roumie CL, Greevy RA, et al. Comparative effectiveness of incident oral antidiabetic drugs on kidney function. *Kidney Int*. 2012;81:698–706.
94. Carrero JJ, Qureshi AR, Axelsson J, et al. Clinical and biochemical implications of low thyroid hormone levels (total and free forms) in euthyroid patients with chronic kidney disease. *J Intern Med*. 2007;262:690–701.
95. Zoccali C, Mallamaci F, Tripepi G, et al. Low triiodothyronine and survival in end-stage renal disease. *Kidney Int*. 2006;70:523–528.
102. Meuwese CL, Carrero JJ, Cabezas-Rodriguez I, et al. Nonthyroidal illness: a risk factor for coronary calcification and arterial stiffness in patients undergoing peritoneal dialysis? *J Intern Med*. 2013;274:584–593.
113. Mak RH, Cheung WW, Roberts CT Jr. The growth hormone-insulin-like growth factor-I axis in chronic kidney disease. *Growth Horm IGF Res*. 2008;18:17–25.
114. Mahesh S, Kaskel F. Growth hormone axis in chronic kidney disease. *Pediatr Nephrol*. 2008;23:41–48.
124. Ding H, Gao XL, Hirschberg R, et al. Impaired actions of insulin-like growth factor I on protein synthesis and degradation in skeletal muscle of rats with chronic renal failure: evidence for a postreceptor defect. *J Clin Invest*. 1996;97:1064–1075.
126. Sun DF, Zheng Z, Tummala P, et al. Chronic uremia attenuates growth hormone-induced signal transduction in skeletal muscle. *J Am Soc Nephrol*. 2004;15:2630–2636.
127. Wang H, Casaburi R, Taylor WE, et al. Skeletal muscle mRNA for IGF-IEa, IGF-II, and IGF-I receptor is decreased in sedentary chronic hemodialysis patients. *Kidney Int*. 2005;68:352–361.
130. Garibotto G, Russo R, Sofia A, et al. Effects of uremia and inflammation on growth hormone resistance in patients with chronic kidney diseases. *Kidney Int*. 2008;74:937–945.
133. Fine RN, Ho M, Tejani A, et al. Adverse events with rhGH treatment of patients with chronic renal insufficiency and end-stage renal disease. *J Pediatr*. 2003;142:539–545.
136. Feldt-Rasmussen B, Lange M, Sulowicz W, et al. Growth hormone treatment during hemodialysis in a randomized trial improves nutrition, quality of life, and cardiovascular risk. *J Am Soc Nephrol*. 2007;18:2161–2171.
137. Guebre-Egziabher F, Juillard L, Boirie Y, et al. Short-term administration of a combination of recombinant growth hormone and insulin-like growth factor-I induces anabolism in maintenance hemodialysis. *J Clin Endocrinol Metab*. 2009;94:2299–2305.
152. Carrero JJ, Kyriazis J, Sonmez A, et al. Prolactin levels, endothelial dysfunction, and the risk of cardiovascular events and mortality in patients with CKD. *Clin J Am Soc Nephrol*. 2012;7:207–215.
162. Mejia-Rodriguez O, Herrera-Abarca JE, Ceballos-Reyes G, et al. Cardiovascular and renal effects of bromocriptine in diabetic patients with stage 4 chronic kidney disease. *Biomed Res Int*. 2013;2013:104059.
165. Gong R. The renaissance of corticotropin therapy in proteinuric nephropathies. *Nat Rev Nephrol*. 2012;8:122–128.
171. Drechsler C, Ritz E, Tomaschitz A, et al. Aldosterone and cortisol affect the risk of sudden cardiac death in haemodialysis patients. *Eur Heart J*. 2013;34:578–587.
174. Kakiya R, Shoji T, Hayashi T, et al. Decreased serum adrenal androgen dehydroepiandrosterone sulfate and mortality in hemodialysis patients. *Nephrol Dial Transplant*. 2012;27:3915–3922.
179. Anantharaman P, Schmidt RJ. Sexual function in chronic kidney disease. *Adv Chronic Kidney Dis*. 2007;14:119–125.
180. Palmer BF. Sexual dysfunction in uremia. *J Am Soc Nephrol*. 1999;10:1381–1388.
189. Anderson GD, Odegard PS. Pharmacokinetics of estrogen and progesterone in chronic kidney disease. *Adv Chronic Kidney Dis*. 2004;11:357–360.
193. Hernandez E, Valera R, Alonzo E, et al. Effects of raloxifene on bone metabolism and serum lipids in postmenopausal women on chronic hemodialysis. *Kidney Int*. 2003;63:2269–2274.
203. Agarwal M, Selvan V, Freedman BI, et al. The relationship between albuminuria and hormone therapy in postmenopausal women. *Am J Kidney Dis*. 2005;45:1019–1025.
211. Carrero JJ, Stenvinkel P. The vulnerable man: impact of testosterone deficiency on the uraemic phenotype. *Nephrol Dial Transplant*. 2012;27:4030–4041.
212. Carrero JJ, Qureshi AR, Nakashima A, et al. Prevalence and clinical implications of testosterone deficiency in men with end-stage renal disease. *Nephrol Dial Transplant*. 2011;26:184–190.
214. Palmer BF. Outcomes associated with hypogonadism in men with chronic kidney disease. *Adv Chronic Kidney Dis*. 2004;11:342–347.
218. Carrero JJ, Qureshi AR, Parini P, et al. Low serum testosterone increases mortality risk among male dialysis patients. *J Am Soc Nephrol*. 2009;20:613–620.
221. Johansen KL, Painter PL, Sakkas GK, et al. Effects of resistance exercise training and nandrolone decanoate on body composition and muscle function among patients who receive hemodialysis: a randomized, controlled trial. *J Am Soc Nephrol*. 2006;17:2307–2314.
222. Johansen KL, Mulligan K, Schambelan M. Anabolic effects of nandrolone decanoate in patients receiving dialysis: a randomized controlled trial. *JAMA*. 1999;281:1275–1281.
226. Carrero JJ, Barany P, Yilmaz MI, et al. Testosterone deficiency is a cause of anemia and reduced responsiveness to erythropoiesis stimulating agents in men with chronic kidney disease. *Nephrol Dial Transplant*. 2012;27:709–715.

254. Kidney Disease: Improving Global Outcomes (KDIGO) CKD-MBD Work Group. KDIGO clinical practice guideline for the diagnosis, evaluation, prevention, and treatment of chronic kidney Disease-Mineral and bone disorder (CKD-MBD). *Kidney Int Suppl.* 2009;113:S1–S130.
261. Agarwal R. Vitamin D, proteinuria, diabetic nephropathy, and progression of CKD. *Clin J Am Soc Nephrol.* 2009;4:1523–1528.
265. Garrett G, Sardiwal S, Lamb EJ, et al. PTH—a particularly tricky hormone: why measure it at all in kidney patients? *Clin J Am Soc Nephrol.* 2013;8:299–312.
268. Ravani P, Malberti F, Tripepi G, et al. Vitamin D levels and patient outcome in chronic kidney disease. *Kidney Int.* 2009;75:88–95.
279. Duranton F, Rodriguez-Ortiz ME, Duny Y, et al. Vitamin D treatment and mortality in chronic kidney disease: a systematic review and meta-analysis. *Am J Nephrol.* 2013;37:239–248.

REFERENCES

- DeFronzo RA, Smith D, Alvestrand A. Insulin action in uremia. *Kidney Int Suppl*. 1983;16:S102–S114.
- Mak RH, Bettinelli A, Turner C, et al. The influence of hyperparathyroidism on glucose metabolism in uremia. *J Clin Endocrinol Metab*. 1985;60:229–233.
- Mak RH. Intravenous 1,25 dihydroxycholecalciferol corrects glucose intolerance in hemodialysis patients. *Kidney Int*. 1992;41:1049–1054.
- DeFronzo RA, Tobin JD, Andres R. Glucose clamp technique: a method for quantifying insulin secretion and resistance. *Am J Physiol*. 1979;237:E214–E223.
- Jia T, Huang X, Qureshi AR, et al. Validation of insulin sensitivity surrogate indices and prediction of clinical outcomes in individuals with and without impaired renal function. *Kidney Int*. 2014;86:383–391.
- Reaven GM. Banting lecture 1988: role of insulin resistance in human disease. *Diabetes*. 1988;37:1595–1607.
- DeFronzo RA, Alvestrand A, Smith D, et al. Insulin resistance in uremia. *J Clin Invest*. 1981;67:563–568.
- Friedman JE, Dohm GL, Elton CW, et al. Muscle insulin resistance in uremic humans: glucose transport, glucose transporters, and insulin receptors. *Am J Physiol*. 1991;261:E87–E94.
- Guthoff M, Wagner R, Vosseler D, et al. Impact of end-stage renal disease on glucose metabolism—a matched cohort analysis. *Nephrol Dial Transplant*. 2017;32(4):670–676.
- Marette A. Mediators of cytokine-induced insulin resistance in obesity and other inflammatory settings. *Curr Opin Clin Nutr Metab Care*. 2002;5:377–383.
- Dion F, Dumayne C, Henley N, et al. Mechanism of insulin resistance in a rat model of kidney disease and the risk of developing type 2 diabetes. *PLoS ONE*. 2017;12(5):e0176650.
- Contreras I, Caro JF, Avelledo L, et al. In chronic uremia, insulin activates receptor kinase but not pyruvate dehydrogenase. *Nephron*. 1992;61:77–81.
- Bailey JL, Zheng B, Hu Z, et al. Chronic kidney disease causes defects in signaling through the insulin receptor substrate/phosphatidylinositol 3-kinase/Akt pathway: implications for muscle atrophy. *J Am Soc Nephrol*. 2006;17:1388–1394.
- Kawazoe Y, Naka T, Fujimoto M, et al. Signal transducer and activator of transcription (STAT)-induced STAT inhibitor 1 (SSI-1)/suppressor of cytokine signaling 1 (SOCS1) inhibits insulin signal transduction pathway through modulating insulin receptor substrate 1 (IRS-1) phosphorylation. *J Exp Med*. 2001;193:263–269.
- Deger SM, Wang P, Fissell R, et al. Tissue sodium accumulation and peripheral insulin sensitivity in maintenance hemodialysis patients. *J Cachexia Sarcopenia Muscle*. 2017;8(3):500–507.
- Mak RH. 1,25-Dihydroxyvitamin D₃ corrects insulin and lipid abnormalities in uremia. *Kidney Int*. 1998;53:1353–1357.
- Chiu KC, Chu A, Go VL, et al. Hypovitaminosis D is associated with insulin resistance and beta cell dysfunction. *Am J Clin Nutr*. 2004;79:820–825.
- Trirógoff ML, Shintani A, Himmelfarb J, et al. Body mass index and fat mass are the primary correlates of insulin resistance in nondiabetic stage 3–4 chronic kidney disease patients. *Am J Clin Nutr*. 2007;86:1642–1648.
- Sanchez-Lozada LG, Tapia E, Jimenez A, et al. Fructose-induced metabolic syndrome is associated with glomerular hypertension and renal microvascular damage in rats. *Am J Physiol Renal Physiol*. 2007;292:F423–F429.
- Cirillo P, Gersch MS, Mu W, et al. Ketohexokinase-dependent metabolism of fructose induces proinflammatory mediators in proximal tubular cells. *J Am Soc Nephrol*. 2009;20:545–553.
- Church JM, Hill GL. Impaired glucose metabolism in surgical patients improved by intravenous nutrition: assessment by the euglycemic-hyperinsulinemic clamp. *Metabolism*. 1988;37:505–509.
- Fève B, Bastard JP. The role of interleukins in insulin resistance and type 2 diabetes mellitus. *Nat Rev Endocrinol*. 2009;5:305–311.
- Beddhu S, Kimmel PL, Ramkumar N, et al. Associations of metabolic syndrome with inflammation in CKD: results from the Third National Health and Nutrition Examination Survey (NHANES III). *Am J Kidney Dis*. 2005;46:577–586.
- Park JT, Chang TI, Kim DK, et al. Association of white blood cell count with metabolic syndrome in patients undergoing peritoneal dialysis. *Metabolism*. 2009;58:1379–1385.
- Lofgren P, van Harmelen V, Reynisdottir S, et al. Secretion of tumor necrosis factor- α shows a strong relationship to insulin-stimulated glucose transport in human adipose tissue. *Diabetes*. 2000;49:688–692.
- Larsen CM, Faulenbach M, Vaag A, et al. Interleukin-1-receptor antagonist in type 2 diabetes mellitus. *N Engl J Med*. 2007;356:1517–1526.
- Rieusset J, Bouzakri K, Chevillotte E, et al. Suppressor of cytokine signaling 3 expression and insulin resistance in skeletal muscle of obese and type 2 diabetic patients. *Diabetes*. 2004;53:2232–2241.
- Raj DS, Dominic EA, Pai A, et al. Skeletal muscle, cytokines, and oxidative stress in end-stage renal disease. *Kidney Int*. 2005;68:2338–2344.
- D'Apolito M, Du X, Zong H, et al. Urea-induced ROS generation causes insulin resistance in mice with chronic renal failure. *J Clin Invest*. 2010;120:203–213.
- Mak RH. Correction of anemia by erythropoietin reverses insulin resistance and hyperinsulinemia in uremia. *Am J Physiol*. 1996;270:F839–F844.
- Spaia S, Pangalos M, Askepidis N, et al. Effect of short-term rHuEPO treatment on insulin resistance in haemodialysis patients. *Nephron*. 2000;84:320–325.
- Abe M, Okada K, Soma M, et al. Relationship between insulin resistance and erythropoietin responsiveness in hemodialysis patients. *Clin Nephrol*. 2011;75:49–58.
- Cheng SC, Chu TS, Huang KY, et al. Association of hypertriglyceridemia and insulin resistance in uremic patients undergoing CAPD. *Perit Dial Int*. 2001;21:282–289.
- Allegra V, Mengozzi G, Martimbianco L, et al. Glucose-induced insulin secretion in uremia: effects of aminophylline infusion and glucose loads. *Kidney Int*. 1990;38:1146–1150.
- Delarue J, Maingourd C. Acute metabolic effects of dialysis fluids during CAPD. *Am J Kidney Dis*. 2001;37(suppl 2):S103–S107.
- Amici G, Orrasch M, Da Rin G, et al. Hyperinsulinism reduction associated with icodextrin treatment in continuous ambulatory peritoneal dialysis patients. *Adv Perit Dial*. 2001;17:80–83.
- Furuya R, Odamaki M, Kumagai H, et al. Beneficial effects of icodextrin on plasma level of adipocytokines in peritoneal dialysis patients. *Nephrol Dial Transplant*. 2006;21:494–498.
- Kurella M, Lo JC, Chertow GM. Metabolic syndrome and the risk for chronic kidney disease among nondiabetic adults. *J Am Soc Nephrol*. 2005;16:2134–2140.
- Chen J, Muntner P, Hamm LL, et al. The metabolic syndrome and chronic kidney disease in U.S. adults. *Ann Intern Med*. 2004;140:167–174.
- Kobayashi H, Tokudome G, Hara Y, et al. Insulin resistance is a risk factor for the progression of chronic kidney disease. *Clin Nephrol*. 2009;71:643–651.
- Böttiger EP. TGF- β in renal injury and disease. *Semin Nephrol*. 2007;27:309–320.
- Shinohara K, Shoji T, Emoto M, et al. Insulin resistance as an independent predictor of cardiovascular mortality in patients with end-stage renal disease. *J Am Soc Nephrol*. 2002;13:1894–1900.
- He J, Shlipak M, Anderson A, et al. CRIC (Chronic Renal Insufficiency Cohort) Investigators. Risk factors for heart failure in patients with chronic kidney disease: the CRIC (chronic renal insufficiency cohort) study. *J Am Heart Assoc*. 2017;6(5).
- Stenvinkel P, Ottosson-Seeberger A, Alvestrand A. Renal hemodynamics and sodium handling in moderate renal insufficiency: the role of insulin resistance and dyslipidemia. *J Am Soc Nephrol*. 1995;5:1751–1760.
- Liu Z. The renin-angiotensin system and insulin resistance. *Curr Diab Rep*. 2007;7:34–42.
- Olesen P, Nguyen K, Wogensen L, et al. Calcification of human vascular smooth muscle cells: associations with osteoprotegerin expression and acceleration by high-dose insulin. *Am J Physiol Heart Circ Physiol*. 2007;292:H1058–H1064.
- Fadini GP, Pauletto P, Avogaro A, et al. The good and the bad in the link between insulin resistance and vascular calcification. *Atherosclerosis*. 2007;193:241–244.
- Lee SW, Dai G, Hu Z, et al. Regulation of muscle protein degradation: coordinated control of apoptotic and ubiquitin-proteasome systems by phosphatidylinositol 3 kinase. *J Am Soc Nephrol*. 2004;15:1537–1545.
- Pupim LB, Flakoll PJ, Majchrzak KM, et al. Increased muscle protein breakdown in chronic hemodialysis patients with type 2 diabetes mellitus. *Kidney Int*. 2005;68:1857–1865.
- Mustata S, Chan C, Lai V, et al. Impact of an exercise program on arterial stiffness and insulin resistance in hemodialysis patients. *J Am Soc Nephrol*. 2004;15:2713–2718.

51. Yamamoto-Kabasawa K, Hosojima M, Yata Y, et al. Benefits of a 12-week lifestyle modification program including diet and combined aerobic and resistance exercise on albuminuria in diabetic and non-diabetic Japanese populations. *Clin Exp Nephrol*. 2015;19(6):1079–1089. Epub 2015 Mar 8.
52. McFarlane SI, Kumar A, Sowers JR. Mechanisms by which angiotensin-converting enzyme inhibitors prevent diabetes and cardiovascular disease. *Am J Cardiol*. 2003;91:30H–37H.
53. Bosch J, Yusuf S, Gerstein HC, et al. Effect of ramipril on the incidence of diabetes. *N Engl J Med*. 2006;355:1551–1562.
54. Wheeler S, Moore K, Forsberg CW, et al. Mortality among veterans with type 2 diabetes initiating metformin, sulfonylurea or rosiglitazone monotherapy. *Diabetologia*. 2013;56:1934–1943.
55. Gerstein HC, Yusuf S, Bosch J, et al. Effect of rosiglitazone on the frequency of diabetes in patients with impaired glucose tolerance or impaired fasting glucose: a randomised controlled trial. *Lancet*. 2006;368:1096–1105.
56. Ramirez SP, Albert JM, Blayney MJ, et al. Rosiglitazone is associated with mortality in chronic hemodialysis patients. *J Am Soc Nephrol*. 2009;20:1094–1101.
57. Nissen SE, Wolski K. Effect of rosiglitazone on the risk of myocardial infarction and death from cardiovascular causes. *N Engl J Med*. 2007;356:2457–2471.
58. Chan DT, Watts GF, Irish AB, et al. Rosiglitazone does not improve vascular function in subjects with chronic kidney disease. *Nephrol Dial Transplant*. 2011;26:3543–3549.
59. Fjita Y, Inagaki N. Metformin: new preparations and nonglycemic benefits. *Curr Diab Rep*. 2017;17(1):5.
60. Imam TH. Changes in metformin use in chronic kidney disease. *Clin Kidney J*. 2017;10(3):301–304.
61. Salber GJ, Wang Y-B, Lynch JT, et al. Metformin use in practice: compliance with guidelines for patients with diabetes and preserved renal function. *Clin Diabetes*. 2017;35(3):154–161.
62. Hung AM, Roumie CL, Greevy RA, et al. Comparative effectiveness of incident oral antidiabetic drugs on kidney function. *Kidney Int*. 2012;81:698–706.
63. Zinman B, Wanner C, Lachin JM, et al; EMPA-REG OUTCOME Investigators. Empagliflozin, cardiovascular outcomes, and mortality in type 2 diabetes. *N Engl J Med*. 2015;373(22):2117–2128.
64. Wanner C, Inzucchi SE, Lachin JM, et al; EMPA-REG OUTCOME Investigators. Empagliflozin and progression of kidney disease in type 2 diabetes. *N Engl J Med*. 2016;375(4):323–334. Epub 2016 Jun 14.
65. Sato K, Okamura K, Yoshinari M, et al. Reversible primary hypothyroidism and elevated serum iodine level in patients with renal dysfunction. *Acta Endocrinol (Copenh)*. 1992;126:253–259.
66. Ramirez G, Jubiz W, Gutch CF, et al. Thyroid abnormalities in renal failure: a study of 53 patients on chronic hemodialysis. *Ann Intern Med*. 1973;79:500–504.
67. Vanhorebeek I, Langouche L, Van den Berghe G. Endocrine aspects of acute and prolonged critical illness. *Nat Clin Pract Endocrinol Metab*. 2006;2:20–31.
68. Wiederkehr MR, Kalogiros J, Krapf R. Correction of metabolic acidosis improves thyroid and growth hormone axes in haemodialysis patients. *Nephrol Dial Transplant*. 2004;19:1190–1197.
69. Disthabanchong S, Treeruttanawanich A. Oral sodium bicarbonate improves thyroid function in predialysis chronic kidney disease. *Am J Nephrol*. 2010;32:549–556.
70. Warner MH, Beckett GJ. Mechanisms behind the non-thyroidal illness syndrome: an update. *J Endocrinol*. 2010;205:1–13.
71. Torpy DJ, Tsigos C, Lotsikas AJ, et al. Acute and delayed effects of a single-dose injection of interleukin-6 on thyroid function in healthy humans. *Metabolism*. 1998;47:1289–1293.
72. Boelen A, Kwakkel J, Wiersinga WM, et al. Chronic local inflammation in mice results in decreased TRH and type 3 deiodinase mRNA expression in the hypothalamic paraventricular nucleus independently of diminished food intake. *J Endocrinol*. 2006;191:707–714.
73. Wilber JF, Utiger RD. The effect of glucocorticoids on thyrotropin secretion. *J Clin Invest*. 1969;48:2096–2103.
74. Wiersinga WM, Toubert JL. The influence of beta-adrenoceptor blocking agents on plasma thyroxine and triiodothyronine. *J Clin Endocrinol Metab*. 1977;45:293–298.
75. Kaptein EM. Thyroid hormone metabolism and thyroid diseases in chronic renal failure. *Endocr Rev*. 1996;17:45–63.
76. Silverberg DS, Ulan RA, Fawcett DM, et al. Effects of chronic hemodialysis on thyroid function in chronic renal failure. *Can Med Assoc J*. 1973;109:282–286.
77. Amato AA, Santos GM, Neves Fde A. Thyroid hormone action in chronic kidney disease. *Curr Opin Endocrinol Diabetes Obes*. 2008;15:459–465.
78. Lim CF, Bernard BF, de Jong M, et al. A furan fatty acid and indoxyl sulfate are the putative inhibitors of thyroxine hepatocyte transport in uremia. *J Clin Endocrinol Metab*. 1993;76:318–324.
79. Santos GM, Pantoja CJ, Costa ESA, et al. Thyroid hormone receptor binding to DNA and T3-dependent transcriptional activation are inhibited by uremic toxins. *Nucl Recept*. 2005;3:1.
80. Lo JC, Chertow GM, Go AS, et al. Increased prevalence of subclinical and clinical hypothyroidism in persons with chronic kidney disease. *Kidney Int*. 2005;67:1047–1052.
81. Chonchol M, Lippi G, Salvagno G, et al. Prevalence of subclinical hypothyroidism in patients with chronic kidney disease. *Clin J Am Soc Nephrol*. 2008;3:1296–1300.
82. Song SH, Kwak IS, Lee DW, et al. The prevalence of low triiodothyronine according to the stage of chronic kidney disease in subjects with a normal thyroid-stimulating hormone. *Nephrol Dial Transplant*. 2009;24:1534–1538.
83. Ozen KP, Asci G, Gungor O, et al. Nutritional state alters the association between free triiodothyronine levels and mortality in hemodialysis patients. *Am J Nephrol*. 2011;33:305–312.
84. Montenegro J, Gonzalez O, Saracho R, et al. Changes in renal function in primary hypothyroidism. *Am J Kidney Dis*. 1996;27:195–198.
85. Kreisman SH, Hennessey JV. Consistent reversible elevations of serum creatinine levels in severe hypothyroidism. *Arch Intern Med*. 1999;159:79–82.
86. Chou KM, Chiu SY, Chen CH, et al. Correlation of clinical changes with regard to thyroxine replacement therapy in hypothyroid patients - focusing on the change of renal function. *Kidney Blood Press Res*. 2011;34:365–372.
87. Shin DH, Lee MJ, Kim SJ, et al. Preservation of renal function by thyroid hormone replacement therapy in chronic kidney disease patients with subclinical hypothyroidism. *J Clin Endocrinol Metab*. 2012;97:2732–2740.
88. Shin DH, Lee MJ, Lee HS, et al. Thyroid hormone replacement therapy attenuates the decline of renal function in chronic kidney disease patients with subclinical hypothyroidism. *Thyroid*. 2013;23:654–661.
89. Rhee CM, Kim S, Gillen DL. Association of thyroid functional disease with mortality in a national cohort of incident hemodialysis patients. *J Clin Endocrinol Metab*. 2015;100(4):1386–1395. Epub 2015 Jan 29.
90. Rhee CM, Alexander EK, Bhan I, et al. Hypothyroidism and mortality among dialysis patients. *Clin J Am Soc Nephrol*. 2013;8(4):593–601. Epub 2012 Dec 20.
91. Rhee CM, Ravel VA, Streja E, et al. Thyroid functional disease and mortality in a national peritoneal dialysis cohort. *J Clin Endocrinol Metab*. 2016;101(11):4054–4061. Epub 2016 Aug 15.
92. Rhee CM, You AS, Nguyen DV, et al. Thyroid status and mortality in a prospective hemodialysis cohort. *J Clin Endocrinol Metab*. 2017;102(5):1568–1577. doi:10.1210/jc.2016-3616.
93. Rhee CM, Chen Y, You AS, et al. Thyroid status, quality of life, and mental health in patients on hemodialysis. *Clin J Am Soc Nephrol*. 2017;12(8):1274–1283. Epub 2017 Jul 13.
94. Carrero JJ, Qureshi AR, Axelsson J, et al. Clinical and biochemical implications of low thyroid hormone levels (total and free forms) in euthyroid patients with chronic kidney disease. *J Intern Med*. 2007;262:690–701.
95. Zoccali C, Mallamaci F, Tripepi G, et al. Low triiodothyronine and survival in end-stage renal disease. *Kidney Int*. 2006;70:523–528.
96. Enia G, Panuccio V, Cutrupi S, et al. Subclinical hypothyroidism is linked to micro-inflammation and predicts death in continuous ambulatory peritoneal dialysis. *Nephrol Dial Transplant*. 2007;22:538–544.
97. Rhee CM, Alexander EK, Bhan I, et al. Hypothyroidism and mortality among dialysis patients. *Clin J Am Soc Nephrol*. 2013;8:593–601.
98. Meuwese CL, Dekker FW, Lindholm B, et al. Baseline levels and trimestral variation of triiodothyronine and thyroxine and their association with mortality in maintenance hemodialysis patients. *Clin J Am Soc Nephrol*. 2012;7:131–138.

99. Xu H, Brusselaers N, Lindholm B, et al. Thyroid function test derangements and mortality in dialysis patients: a systematic review and Meta-analysis. *Am J Kidney Dis*. 2016;68(6):923–932. Epub 2016 Sep 2.
100. Zoccali C, Benedetto F, Mallamaci F, et al. Low triiodothyronine and cardiomyopathy in patients with end-stage renal disease. *J Hypertens*. 2006;24:2039–2046.
101. Kang EW, Nam JY, Yoo TH, et al. Clinical implications of subclinical hypothyroidism in continuous ambulatory peritoneal dialysis patients. *Am J Nephrol*. 2008;28:908–913.
102. Meuwese CL, Carrero JJ, Cabezas-Rodriguez I, et al. Nonthyroidal illness: a risk factor for coronary calcification and arterial stiffness in patients undergoing peritoneal dialysis? *J Intern Med*. 2013;274:584–593.
103. Tatar E, Kircelli F, Asci G, et al. Associations of triiodothyronine levels with carotid atherosclerosis and arterial stiffness in hemodialysis patients. *Clin J Am Soc Nephrol*. 2011;6:2240–2246.
104. Yilmaz MI, Sonmez A, Karaman M, et al. Low triiodothyronine alters flow-mediated vasodilatation in advanced nondiabetic kidney disease. *Am J Nephrol*. 2011;33:25–32.
105. Tatar E, Sezis Demirci M, Kircelli F, et al. The association between thyroid hormones and arterial stiffness in peritoneal dialysis patients. *Int Urol Nephrol*. 2012;44:601–606.
106. Rotondi M, Netti GS, Rosati A, et al. Pretransplant serum FT3 levels in kidney graft recipients are useful for identifying patients with higher risk for graft failure. *Clin Endocrinol (Oxf)*. 2008;68:220–225.
107. Lim VS, Tsalikian E, Flanagan MJ. Augmentation of protein degradation by L-triiodothyronine in uremia. *Metabolism*. 1989;38(12):1210–1215.
108. Disthabanchong S, Treeruttanawanich A. Oral sodium bicarbonate improves thyroid function in predialysis chronic kidney disease. *Am J Nephrol*. 2010;32(6):549–556. Epub 2010 Nov 2.
109. Vidart J, Wajner SM, Leite RS, et al. N-acetylcysteine administration prevents nonthyroidal illness syndrome in patients with acute myocardial infarction: a randomized clinical trial. *J Clin Endocrinol Metab*. 2014;99(12):4537–4545.
110. Barcza Stockler-Pinto M, Carrero JJ, De Carvalho Cardoso Weide L, et al. Effect of selenium supplementation via Brazil nut (*Bertholletia Excelsa*, Hbk) on thyroid hormones levels in hemodialysis patients: a pilot study. *Nutr Hosp*. 2015;32(4):1808–1812.
111. Bommer C, Werle E, Walter-Sack I, et al. D-thyroxine reduces lipoprotein(a) serum concentration in dialysis patients. *J Am Soc Nephrol*. 1998;9(1):90–96.
112. Rhee CM, Alexander EK, Bhan I, et al. Hypothyroidism and mortality among dialysis patients. *Clin J Am Soc Nephrol*. 2013;8(4):593–601. Epub 2012 Dec 20.
113. Mak RH, Cheung WW, Roberts CT Jr. The growth hormone-insulin-like growth factor-I axis in chronic kidney disease. *Growth Horm IGF Res*. 2008;18:17–25.
114. Mahesh S, Kaskel F. Growth hormone axis in chronic kidney disease. *Pediatr Nephrol*. 2008;23:41–48.
115. Feldt-Rasmussen B, El Nahas M. Potential role of growth factors with particular focus on growth hormone and insulin-like growth factor-I in the management of chronic kidney disease. *Semin Nephrol*. 2009;29:50–58.
116. Jia T, Gama Axelsson T, Heimbürger O, et al. IGF-I and survival in ESRD. *Clin J Am Soc Nephrol*. 2014;9:120–127.
117. Miller SB, Rabkin R. The use of growth factors to increase glomerular filtration rate in chronic renal failure patients. *Curr Opin Nephrol Hypertens*. 1997;6:401–404.
118. Miller SB, Moulton M, O'Shea M, et al. Effects of IGF-I on renal function in end-stage chronic renal failure. *Kidney Int*. 1994;46:201–207.
119. Vijayan A, Franklin SC, Behrend T, et al. Insulin-like growth factor I improves renal function in patients with end-stage chronic renal failure. *Am J Physiol*. 1999;276:R929–R934.
120. Bach LA, Hale LJ. Insulin-like growth factors and kidney disease. *Am J Kidney Dis*. 2015;65(2):327–336. Epub 2014 Aug 21.
121. Doi T, Striker LJ, Gibson CC, et al. Glomerular lesions in mice transgenic for growth hormone and insulinlike growth factor-I. I: relationship between increased glomerular size and mesangial sclerosis. *Am J Pathol*. 1990;137:541–552.
122. Doi T, Striker LJ, Quaife C, et al. Progressive glomerulosclerosis develops in transgenic mice chronically expressing growth hormone and growth hormone releasing factor but not in those expressing insulinlike growth factor-I. *Am J Pathol*. 1988;131:398–403.
123. Wuhl E, Schaefer F. Effects of growth hormone in patients with chronic renal failure: experience in children and adults. *Horm Res*. 2002;58(suppl 3):35–38.
124. Ding H, Gao XL, Hirschberg R, et al. Impaired actions of insulin-like growth factor I on protein synthesis and degradation in skeletal muscle of rats with chronic renal failure: evidence for a postreceptor defect. *J Clin Invest*. 1996;97:1064–1075.
125. Rabkin R, Sun DF, Chen Y, et al. Growth hormone resistance in uremia, a role for impaired JAK/STAT signaling. *Pediatr Nephrol*. 2005;20:313–318.
126. Sun DF, Zheng Z, Tummala P, et al. Chronic uremia attenuates growth hormone-induced signal transduction in skeletal muscle. *J Am Soc Nephrol*. 2004;15:2630–2636.
127. Wang H, Casaburi R, Taylor WE, et al. Skeletal muscle mRNA for IGF-IEa, IGF-II, and IGF-I receptor is decreased in sedentary chronic hemodialysis patients. *Kidney Int*. 2005;68:352–361.
128. Tonshoff B, Powell DR, Zhao D, et al. Decreased hepatic insulin-like growth factor (IGF)-I and increased IGF binding protein-1 and -2 gene expression in experimental uremia. *Endocrinology*. 1997;138:938–946.
129. Blum WF, Ranke MB, Kietzmann K, et al. Growth hormone resistance and inhibition of somatomedin activity by excess of insulin-like growth factor binding protein in uraemia. *Pediatr Nephrol*. 1991;5:539–544.
130. Garibotto G, Russo R, Sofia A, et al. Effects of uremia and inflammation on growth hormone resistance in patients with chronic kidney diseases. *Kidney Int*. 2008;74:937–945.
131. Lewy JE. Treatment of children in the U.S. with end-stage renal disease (ESRD). *Med Arh*. 2001;55:201–202.
132. Haffner D, Schaefer F, Nissel R, et al. Effect of growth hormone treatment on the adult height of children with chronic renal failure: German Study Group for growth hormone treatment in chronic renal failure. *N Engl J Med*. 2000;343:923–930.
133. Fine RN, Ho M, Tejani A, et al. Adverse events with rhGH treatment of patients with chronic renal insufficiency and end-stage renal disease. *J Pediatr*. 2003;142:539–545.
134. Hokken-Koelega A, Mulder P, De Jong R, et al. Long-term effects of growth hormone treatment on growth and puberty in patients with chronic renal insufficiency. *Pediatr Nephrol*. 2000;14:701–706.
135. Rees L. Growth hormone therapy in children with CKD after more than two decades of practice. *Pediatr Nephrol*. 2016;31(9):1421–1435. Epub 2015 Sep 14.
136. Feldt-Rasmussen B, Lange M, Sulowicz W, et al. Growth hormone treatment during hemodialysis in a randomized trial improves nutrition, quality of life, and cardiovascular risk. *J Am Soc Nephrol*. 2007;18:2161–2171.
137. Guebre-Egziabher F, Juillard L, Boirie Y, et al. Short-term administration of a combination of recombinant growth hormone and insulin-like growth factor-I induces anabolism in maintenance hemodialysis. *J Clin Endocrinol Metab*. 2009;94:2299–2305.
138. Kopple JD, Brunori G, Leiserowitz M, et al. Growth hormone induces anabolism in malnourished maintenance haemodialysis patients. *Nephrol Dial Transplant*. 2005;20:952–958.
139. Svensson J, Bengtsson BA. Safety aspects of GH replacement. *Eur J Endocrinol*. 2009;161(suppl 1):S65–S74.
140. Zheng Z, Sun DF, Tummala P, et al. Cardiac resistance to growth hormone in uremia. *Kidney Int*. 2005;67:858–866.
141. Takala J, Ruokonen E, Webster NR, et al. Increased mortality associated with growth hormone treatment in critically ill adults. *N Engl J Med*. 1999;341:785–792.
142. Mekala KC, Tritos NA. Effects of recombinant human growth hormone therapy in obesity in adults: a meta analysis. *J Clin Endocrinol Metab*. 2009;94:130–137.
143. Nissel R, Fischer DC, Puhlmann A, et al. Short-term growth hormone treatment and microcirculation: effects in patients with chronic kidney disease. *Microvasc Res*. 2009;78:246–252.
144. Feld S, Hirschberg R. Growth hormone, the insulin-like growth factor system, and the kidney. *Endocr Rev*. 1996;17:423–480.
145. Hou SH, Grossman S, Molitch ME. Hyperprolactinemia in patients with renal insufficiency and chronic renal failure requiring hemodialysis or chronic ambulatory peritoneal dialysis. *Am J Kidney Dis*. 1985;6:245–249.
146. Cowden EA, Ratcliffe WA, Ratcliffe JG, et al. Hyperprolactinaemia in renal disease. *Clin Endocrinol (Oxf)*. 1978;9:241–248.

147. Yavuz D, Topcu G, Ozener C, et al. Macroprolactin does not contribute to elevated levels of prolactin in patients on renal replacement therapy. *Clin Endocrinol (Oxf)*. 2005;63:520–524.
148. Veldhuis JD, Iranmanesh A, Wilkowski MJ, et al. Neuroendocrine alterations in the somatotrophic and lactotrophic axes in uremic men. *Eur J Endocrinol*. 1994;131:489–498.
149. Bommer J, Ritz E, del Pozo E, et al. Improved sexual function in male haemodialysis patients on bromocriptine. *Lancet*. 1979;2(8141):496–497.
150. Schaefer RM, Kokot F, Kuerner B, et al. Normalization of serum prolactin levels in hemodialysis patients on recombinant human erythropoietin. *Int J Artif Organs*. 1989;12:445–449.
151. Schaefer RM, Kokot F, Wernze H, et al. Improved sexual function in hemodialysis patients on recombinant erythropoietin: a possible role for prolactin. *Clin Nephrol*. 1989;31:1–5.
152. Carrero JJ, Kyriazis J, Sonmez A, et al. Prolactin levels, endothelial dysfunction, and the risk of cardiovascular events and mortality in patients with CKD. *Clin J Am Soc Nephrol*. 2012;7:207–215.
153. Francis K, Lewis BM, Akatsu H, et al. Complement C3a receptors in the pituitary gland: a novel pathway by which an innate immune molecule releases hormones involved in the control of inflammation. *FASEB J*. 2003;17:2266–2268.
154. Sauro MD, Buckley AR, Russell DH, et al. Prolactin stimulation of protein kinase C activity in rat aortic smooth muscle. *Life Sci*. 1989;44(23):1787–1792.
155. Aziz MM, Ishihara S, Rumi MA, et al. Prolactin induces MFG-E8 production in macrophages via transcription factor C/EBPbeta-dependent pathway. *Apoptosis*. 2008;13:609–620.
156. Bresson JL, Jeay S, Gagnerault MC, et al. Growth hormone (GH) and prolactin receptors in human peripheral blood mononuclear cells: relation with age and GH-binding protein. *Endocrinology*. 1999;140:3203–3209.
157. Sun R, Li AL, Wei HM, et al. Expression of prolactin receptor and response to prolactin stimulation of human NK cell lines. *Cell Res*. 2004;14:67–73.
158. Molinari C, Grossini E, Mary DA, et al. Prolactin induces regional vasoconstriction through the beta2-adrenergic and nitric oxide mechanisms. *Endocrinology*. 2007;148:4080–4090.
159. Mejia-Rodriguez O, Alvarez-Aguilar C, Vega-Gomez HE, et al. Bromocriptine induces regression of left ventricular hypertrophy in peritoneal dialysis patients. *Proc West Pharmacol Soc*. 2005;48:122–125.
160. Mejia-Rodriguez O, Alvarez-Aguilar C, Ledesma-Ramirez M, et al. Therapeutic effect of bromocriptine together with the established treatment for hypertension in patients undergoing peritoneal dialysis. *Proc West Pharmacol Soc*. 2004;47:122–124.
161. Degli Esposti E, Sturani A, Santoro A, et al. Effect of bromocriptine treatment on prolactin, noradrenaline and blood pressure in hypertensive haemodialysis patients. *Clin Sci*. 1985;69:51–56.
162. Mejia-Rodriguez O, Herrera-Abarca JE, Ceballos-Reyes G, et al. Cardiovascular and renal effects of bromocriptine in diabetic patients with stage 4 chronic kidney disease. *Biomed Res Int*. 2013;2013:104059.
163. N'Gankam V, Uehlinger D, Dick B, et al. Increased cortisol metabolites and reduced activity of 11beta-hydroxysteroid dehydrogenase in patients on hemodialysis. *Kidney Int*. 2002;61:1859–1866.
164. Si J, Ge Y, Zhuang S, et al. Adrenocorticotrophic hormone ameliorates acute kidney injury by steroidogenic-dependent and -independent mechanisms. *Kidney Int*. 2013;83:635–646.
165. Gong R. The renaissance of corticotropin therapy in proteinuric nephropathies. *Nat Rev Nephrol*. 2012;8:122–128.
166. Iwashima F, Yoshimoto T, Minami I, et al. Aldosterone induces superoxide generation via Rac1 activation in endothelial cells. *Endocrinology*. 2008;149:1009–1014.
167. Leopold JA, Dam A, Maron BA, et al. Aldosterone impairs vascular reactivity by decreasing glucose-6-phosphate dehydrogenase activity. *Nat Med*. 2007;13:189–197.
168. Garnier A, Bendall JK, Fuchs S, et al. Cardiac specific increase in aldosterone production induces coronary dysfunction in aldosterone synthase-transgenic mice. *Circulation*. 2004;110:1819–1825.
169. Endemann DH, Touyz RM, Iglarz M, et al. Eplerenone prevents salt-induced vascular remodeling and cardiac fibrosis in stroke-prone spontaneously hypertensive rats. *Hypertension*. 2004;43:1252–1257.
170. N'Gankam V, Uehlinger D, Dick B, et al. Increased cortisol metabolites and reduced activity of 11beta-hydroxysteroid dehydrogenase in patients on hemodialysis. *Kidney Int*. 2002;61:1859–1866.
171. Drechsler C, Ritz E, Tomaschitz A, et al. Aldosterone and cortisol affect the risk of sudden cardiac death in haemodialysis patients. *Eur Heart J*. 2013;34:578–587.
172. Tomaszewski M, Charchar FJ, Maric C, et al. Inverse associations between androgens and renal function: the Young Men Cardiovascular Association (YMCA) study. *Am J Hypertens*. 2009;22:100–105.
173. Fukui M, Kitagawa Y, Kamiuchi K, et al. Low serum dehydroepiandrosterone sulfate concentration is a predictor for deterioration of urinary albumin excretion in male patients with type 2 diabetes. *Diabetes Res Clin Pract*. 2006;73:47–50.
174. Kakiya R, Shoji T, Hayashi T, et al. Decreased serum adrenal androgen dehydroepiandrosterone sulfate and mortality in hemodialysis patients. *Nephrol Dial Transplant*. 2012;27:3915–3922.
175. Hsu HJ, Yen CH, Chen CK, et al. Low plasma DHEA-S increases mortality risk among male hemodialysis patients. *Exp Gerontol*. 2012;47:950–957.
176. Kawano H, Yasue H, Kitagawa A, et al. Dehydroepiandrosterone supplementation improves endothelial function and insulin sensitivity in men. *J Clin Endocrinol Metab*. 2003;88:3190–3195.
177. Altman R, Motton DD, Kota RS, et al. Inhibition of vascular inflammation by dehydroepiandrosterone sulfate in human aortic endothelial cells: roles of PPARalpha and NF-kappaB. *Vascul Pharmacol*. 2008;48:76–84.
178. Lim VS, Henriquez C, Sievertsen G, et al. Ovarian function in chronic renal failure: evidence suggesting hypothalamic anovulation. *Ann Intern Med*. 1980;93:21–27.
179. Anantharaman P, Schmidt RJ. Sexual function in chronic kidney disease. *Adv Chronic Kidney Dis*. 2007;14:119–125.
180. Palmer BF. Sexual dysfunction in uremia. *J Am Soc Nephrol*. 1999;10:1381–1388.
181. Matuszkiewicz-Rowinska J, Skorzevska K, Radowski S, et al. Endometrial morphology and pituitary-gonadal axis dysfunction in women of reproductive age undergoing chronic haemodialysis—a multicentre study. *Nephrol Dial Transplant*. 2004;19:2074–2077.
182. Weisinger JR, Bellorin-Font E. Outcomes associated with hypogonadism in women with chronic kidney disease. *Adv Chronic Kidney Dis*. 2004;11:361–370.
183. Nappi RE, Wawra K, Schmitt S. Hypoactive sexual desire disorder in postmenopausal women. *Gynecol Endocrinol*. 2006;22:318–323.
184. Burrowes JD, Bluestone PA, Wang J, et al. The effects of moderate doses of megestrol acetate on nutritional status and body composition in a hemodialysis patient. *J Ren Nutr*. 1999;9:89–94.
185. Bocanuso JA, Hutton M, McAllister B. The effects of megestrol acetate on nutritional parameters in a dialysis population. *J Ren Nutr*. 2000;10:36–43.
186. Rammohan M, Kalantar-Zadeh K, Liang A, et al. Megestrol acetate in a moderate dose for the treatment of malnutrition-inflammation complex in maintenance dialysis patients. *J Ren Nutr*. 2005;15:345–355.
187. Monfared A, Heidarzadeh A, Ghaffari M, et al. Effect of megestrol acetate on serum albumin level in malnourished dialysis patients. *J Ren Nutr*. 2009;19:167–171.
188. Carrero JJ, Qureshi AR, Axelsson J, et al. Comparison of nutritional and inflammatory markers in dialysis patients with reduced appetite. *Am J Clin Nutr*. 2007;85:695–701.
189. Anderson GD, Odegard PS. Pharmacokinetics of estrogen and progesterone in chronic kidney disease. *Adv Chronic Kidney Dis*. 2004;11:357–360.
190. Stehman-Breen C, Anderson G, Gibson D, et al. Pharmacokinetics of oral micronized beta-estradiol in postmenopausal women receiving maintenance hemodialysis. *Kidney Int*. 2003;64:290–294.
191. Matuszkiewicz-Rowinska J, Skorzevska K, Radowski S, et al. The benefits of hormone replacement therapy in pre-menopausal women with oestrogen deficiency on haemodialysis. *Nephrol Dial Transplant*. 1999;14:1238–1243.
192. Naves Diaz M, Rodriguez A, Fernandez Martin JL, et al. Effects of estradiol, calcitriol and both treatments combined on bone histomorphometry in rats with chronic kidney disease and ovariectomy. *Bone*. 2007;41:614–619.
193. Hernandez E, Valera R, Alonzo E, et al. Effects of raloxifene on bone metabolism and serum lipids in postmenopausal women on chronic hemodialysis. *Kidney Int*. 2003;63:2269–2274.
194. Holley JL, Schmidt RJ. Hormone replacement therapy in postmenopausal women with end-stage renal disease: a review of the issues. *Semin Dial*. 2001;14:146–149.

195. Simon J, Braunstein G, Nachtigall L, et al. Testosterone patch increases sexual activity and desire in surgically menopausal women with hypoactive sexual desire disorder. *J Clin Endocrinol Metab*. 2005;90:5226–5233.
196. Braunstein GD, Sundwall DA, Katz M, et al. Safety and efficacy of a testosterone patch for the treatment of hypoactive sexual desire disorder in surgically menopausal women: a randomized, placebo-controlled trial. *Arch Intern Med*. 2005;165:1582–1589.
197. Carrero JJ. Gender differences in chronic kidney disease: underpinnings and therapeutic implications. *Kidney Blood Press Res*. 2010;33:383–392.
198. Stringer KD, Komers R, Osman SA, et al. Gender hormones and the progression of experimental polycystic kidney disease. *Kidney Int*. 2005;68:1729–1739.
199. Elliot SJ, Berho M, Korach K, et al. Gender-specific effects of endogenous testosterone: female alpha-estrogen receptor-deficient C57Bl/6J mice develop glomerulosclerosis. *Kidney Int*. 2007;72:464–472.
200. Maric C, Sandberg K, Hinojosa-Laborde C. Glomerulosclerosis and tubulointerstitial fibrosis are attenuated with 17beta-estradiol in the aging dahl salt sensitive rat. *J Am Soc Nephrol*. 2004;15:1546–1556.
201. Catanuto P, Doublier S, Lupia E, et al. 17 beta-estradiol and tamoxifen upregulate estrogen receptor beta expression and control podocyte signaling pathways in a model of type 2 diabetes. *Kidney Int*. 2009;75:1194–1201.
202. Ahmed SB, Hovind P, Parving HH, et al. Oral contraceptives, angiotensin-dependent renal vasoconstriction, and risk of diabetic nephropathy. *Diabetes Care*. 2005;28:1988–1994.
203. Agarwal M, Selvan V, Freedman BI, et al. The relationship between albuminuria and hormone therapy in postmenopausal women. *Am J Kidney Dis*. 2005;45:1019–1025.
204. Szekacs B, Vajo Z, Varbiro S, et al. Postmenopausal hormone replacement improves proteinuria and impaired creatinine clearance in type 2 diabetes mellitus and hypertension. *BJOG*. 2000;107:1017–1021.
205. Monster TB, Janssen WM, de Jong PE, et al. Oral contraceptive use and hormone replacement therapy are associated with microalbuminuria. *Arch Intern Med*. 2001;161:2000–2005.
206. Ahmed SB, Culleton BF, Tonelli M, et al. Oral estrogen therapy in postmenopausal women is associated with loss of kidney function. *Kidney Int*. 2008;74:370–376.
207. Stehman-Breen CO, Gillen D, Gipson D. Prescription of hormone replacement therapy in postmenopausal women with renal failure. *Kidney Int*. 1999;56:2243–2247.
208. Tostain JL, Blanc F. Testosterone deficiency: a common, unrecognized syndrome. *Nat Clin Pract Urol*. 2008;5:388–396.
209. Araujo AB, Esche GR, Kupelian V, et al. Prevalence of symptomatic androgen deficiency in men. *J Clin Endocrinol Metab*. 2007;92:4241–4247.
210. Bello AK, Stenvinkel P, Lin M, et al. Serum testosterone levels and clinical outcomes in male hemodialysis patients. *Am J Kidney Dis*. 2014;63:268–275.
211. Carrero JJ, Stenvinkel P. The vulnerable man: impact of testosterone deficiency on the uraemic phenotype. *Nephrol Dial Transplant*. 2012;27:4030–4041.
212. Carrero JJ, Qureshi AR, Nakashima A, et al. Prevalence and clinical implications of testosterone deficiency in men with end-stage renal disease. *Nephrol Dial Transplant*. 2011;26:184–190.
213. Gungor O, Kircelli F, Carrero JJ, et al. Endogenous testosterone and mortality in male hemodialysis patients: is it the result of aging? *Clin J Am Soc Nephrol*. 2010;5:2018–2023.
214. Palmer BF. Outcomes associated with hypogonadism in men with chronic kidney disease. *Adv Chronic Kidney Dis*. 2004;11:342–347.
215. Schmidt A, Luger A, Horl WH. Sexual hormone abnormalities in male patients with renal failure. *Nephrol Dial Transplant*. 2002;17:368–371.
216. Vita G, Bellinieri G, Trusso A, et al. Uremic autonomic neuropathy studied by spectral analysis of heart rate. *Kidney Int*. 1999;56:232–237.
217. Herbst KL, Bhasin S. Testosterone action on skeletal muscle. *Curr Opin Clin Nutr Metab Care*. 2004;7:271–277.
218. Carrero JJ, Qureshi AR, Parini P, et al. Low serum testosterone increases mortality risk among male dialysis patients. *J Am Soc Nephrol*. 2009;20:613–620.
219. Cobo G, Gallar P, Di Gioia C, et al. Hypogonadism associated with muscle atrophy, physical inactivity and ESA hyporesponsiveness in men undergoing haemodialysis. *Nefrologia*. 2017;37(1):54–60. Epub 2016 Aug 21.
220. Cigarran S, Pousa M, Castro MJ, et al. Endogenous testosterone, muscle strength, and fat-free mass in men with chronic kidney disease. *J Ren Nutr*. 2013;23:e89–e95.
221. Johansen KL, Painter PL, Sakka GK, et al. Effects of resistance exercise training and nandrolone decanoate on body composition and muscle function among patients who receive hemodialysis: a randomized, controlled trial. *J Am Soc Nephrol*. 2006;17:2307–2314.
222. Johansen KL, Mulligan K, Schambelan M. Anabolic effects of nandrolone decanoate in patients receiving dialysis: a randomized controlled trial. *JAMA*. 1999;281:1275–1281.
223. Shahidi NT. Androgens and erythropoiesis. *N Engl J Med*. 1973;289:72–80.
224. Teruel JL, Marcen R, Navarro JF, et al. Evolution of serum erythropoietin after androgen administration to hemodialysis patients: a prospective study. *Nephron*. 1995;70:282–286.
225. Ballal SH, Domoto DT, Polack DC, et al. Androgens potentiate the effects of erythropoietin in the treatment of anemia of end-stage renal disease. *Am J Kidney Dis*. 1991;17:29–33.
226. Carrero JJ, Barany P, Yilmaz MI, et al. Testosterone deficiency is a cause of anemia and reduced responsiveness to erythropoiesis stimulating agents in men with chronic kidney disease. *Nephrol Dial Transplant*. 2012;27:709–715.
227. Lawrence IG, Price DE, Howlett TA, et al. Erythropoietin and sexual dysfunction. *Nephrol Dial Transplant*. 1997;12:741–747.
228. Schaefer F, van Kaick B, Veldhuis JD, et al. Changes in the kinetics and biopotency of luteinizing hormone in hemodialyzed men during treatment with recombinant human erythropoietin. *J Am Soc Nephrol*. 1994;5:1208–1215.
229. de Francisco AL, Fernandez Fresnedo G, Rodrigo E, et al. Past, present and future of erythropoietin use in the elderly. *Int Urol Nephrol*. 2002;33:187–193.
230. Beusterien KM, Nissenson AR, Port FK, et al. The effects of recombinant human erythropoietin on functional health and well-being in chronic dialysis patients. *J Am Soc Nephrol*. 1996;7:763–773.
231. Yilmaz MI, Sonmez A, Qureshi AR, et al. Endogenous testosterone, endothelial dysfunction, and cardiovascular events in men with nondialysis chronic kidney disease. *Clin J Am Soc Nephrol*. 2011;6:1617–1625.
232. Kyriazis J, Tzanakis I, Stylianou K, et al. Low serum testosterone, arterial stiffness and mortality in male haemodialysis patients. *Nephrol Dial Transplant*. 2011;26:2971–2977.
233. Karakitsos D, Patrianakos AP, De Groot E, et al. Androgen deficiency and endothelial dysfunction in men with end-stage kidney disease receiving maintenance hemodialysis. *Am J Nephrol*. 2006;26:536–543.
234. Fukata J, Imura H, Nakao K. Cytokines as mediators in the regulation of the hypothalamic-pituitary-adrenocortical function. *J Endocrinol Invest*. 1994;17:141–155.
235. Jones TH, Kennedy RL. Cytokines and hypothalamic-pituitary function. *Cytokine*. 1993;5:531–538.
236. Malkin CJ, Pugh PJ, Jones RD, et al. The effect of testosterone replacement on endogenous inflammatory cytokines and lipid profiles in hypogonadal men. *J Clin Endocrinol Metab*. 2004;89:3313–3318.
237. Corrales JJ, Almeida M, Burgo R, et al. Androgen-replacement therapy depresses the ex vivo production of inflammatory cytokines by circulating antigen-presenting cells in aging type-2 diabetic men with partial androgen deficiency. *J Endocrinol*. 2006;189:595–604.
238. Kalinchenko SY, Tishova YA, Mskhalaya GJ, et al. Effects of testosterone supplementation on markers of the metabolic syndrome and inflammation in hypogonadal men with the metabolic syndrome: the double-blinded placebo-controlled moscow study. *Clin Endocrinol (Oxf)*. 2010;73:602–612.
239. Singh AB, Norris K, Modi N, et al. Pharmacokinetics of a transdermal testosterone system in men with end stage renal disease receiving maintenance hemodialysis and healthy hypogonadal men. *J Clin Endocrinol Metab*. 2001;86:2437–2445.
240. Johansen KL. Treatment of hypogonadism in men with chronic kidney disease. *Adv Chronic Kidney Dis*. 2004;11:348–356.
241. Canguven O, Aykose G, Albayrak S, et al. Efficacy of testosterone gel in the treatment of erectile dysfunction in hypogonadal hemodialysis patients: a pilot study. *Int J Impot Res*. 2010;22:140–145.
242. Brockenbrough AT, Dittrich MO, Page ST, et al. Transdermal androgen therapy to augment EPO in the treatment of anemia of chronic renal disease. *Am J Kidney Dis*. 2006;47:251–262.
243. Ji H, Menini S, Mok K, et al. Gonadal steroid regulation of renal injury in renal wrap hypertension. *Am J Physiol Renal Physiol*. 2005;288:F513–F520.

244. Cho MH, Jung KJ, Jang HS, et al. Orchiectomy attenuates kidney fibrosis after ureteral obstruction by reduction of oxidative stress in mice. *Am J Nephrol*. 2012;35:7–16.
245. Metcalfe PD, Leslie JA, Campbell MT, et al. Testosterone exacerbates obstructive renal injury by stimulating TNF-alpha production and increasing proapoptotic and profibrotic signaling. *Am J Physiol Endocrinol Metab*. 2008;294:E435–E443.
246. Reckelhoff JF, Zhang H, Srivastava K. Gender differences in development of hypertension in spontaneously hypertensive rats: role of the renin-angiotensin system. *Hypertension*. 2000;35:480–483.
247. Haring R, Nauck M, Volzke H, et al. Low serum testosterone is associated with increased mortality in men with stage 3 or greater nephropathy. *Am J Nephrol*. 2011;33:209–217.
248. Lapi F, Azoulay L, Niazi MT, et al. Androgen deprivation therapy and risk of acute kidney injury in patients with prostate cancer. *JAMA*. 2013;310:289–296.
249. Molinari C, Battaglia A, Grossini E, et al. The effect of testosterone on regional blood flow in prepubertal anaesthetized pigs. *J Physiol*. 2002;543:365–372.
250. Soljancic A, Ruiz AL, Chandrashekar K, et al. Protective role of testosterone in ischemia-reperfusion-induced acute kidney injury. *Am J Physiol Regul Integr Comp Physiol*. 2013;304:R951–R958.
251. Johal M, Levin A. Vitamin D and parathyroid hormone in general populations: understandings in 2009 and applications to chronic kidney disease. *Clin J Am Soc Nephrol*. 2009;4:1508–1514.
252. Holick MF. Vitamin D deficiency. *N Engl J Med*. 2007;357:266–281.
253. Holick MF. Vitamin D status: measurement, interpretation, and clinical application. *Ann Epidemiol*. 2009;19:73–78.
254. Kidney Disease: Improving Global Outcomes (KDIGO) CKD-MBD Work Group. KDIGO clinical practice guideline for the diagnosis, evaluation, prevention, and treatment of Chronic Kidney Disease-Mineral and Bone Disorder (CKD-MBD). *Kidney Int Suppl*. 2009;113:S1–S130.
255. Dusilova-Sulkova S. Vitamin D metabolism and vitamin D traditional and nontraditional, target organs: implications for kidney patients. *J Ren Care*. 2009;35(suppl 1):39–44.
256. Norman AW. From vitamin D to hormone D: fundamentals of the vitamin D endocrine system essential for good health. *Am J Clin Nutr*. 2008;88:491S–499S.
257. Artaza JN, Mehrotra R, Norris KC. Vitamin D and the cardiovascular system. *Clin J Am Soc Nephrol*. 2009;4:1515–1522.
258. Danescu LG, Levy S, Levy J. Vitamin D and diabetes mellitus. *Endocrine*. 2009;35:11–17.
259. Visser M, Deeg DJ, Lips P. Low vitamin D and high parathyroid hormone levels as determinants of loss of muscle strength and muscle mass (sarcopenia): the Longitudinal Aging Study Amsterdam. *J Clin Endocrinol Metab*. 2003;88:5766–5772.
260. Molina P, Carrero JJ, Bover J, et al; European Renal Nutrition (ERN) and Chronic Kidney Disease-Mineral and Bone Disorder (CKD-MBD) Working Groups of the European Renal Association-European Dialysis Transplant Association (ERA-EDTA). Vitamin D, a modulator of musculoskeletal health in chronic kidney disease. *J Cachexia Sarcopenia Muscle*. 2017;8(5):686–701. Epub 2017 Jul 3.
261. Agarwal R. Vitamin D, proteinuria, diabetic nephropathy, and progression of CKD. *Clin J Am Soc Nephrol*. 2009;4:1523–1528.
262. Li YC. Vitamin D and diabetic nephropathy. *Curr Diab Rep*. 2008;8:464–469.
263. Lee D, Rutter M, O'Neill T, et al. Vitamin D, parathyroid hormone and the metabolic syndrome in middle-aged and older European men. *Eur J Endocrinol*. 2009;161:947–954.
264. Jia T, Qureshi AR, Brandenburg V, et al. Determinants of fibroblast growth factor-23 and parathyroid hormone variability in dialysis patients. *Am J Nephrol*. 2013;37:462–471.
265. Garrett G, Sardiwal S, Lamb EJ, et al. PTH—a particularly tricky hormone: why measure it at all in kidney patients? *Clin J Am Soc Nephrol*. 2013;8:299–312.
266. Smith ER, McMahon LP, Holt SG. Fibroblast growth factor 23. *Ann Clin Biochem*. 2014;51(Pt 2):203–227.
267. Kalantar-Zadeh K, Kovesdy CP. Clinical outcomes with active versus nutritional vitamin D compounds in chronic kidney disease. *Clin J Am Soc Nephrol*. 2009;4:1529–1539.
268. Ravani P, Malberti F, Tripepi G, et al. Vitamin D levels and patient outcome in chronic kidney disease. *Kidney Int*. 2009;75:88–95.
269. Gutierrez OM, Isakova T, Andress DL, et al. Prevalence and severity of disordered mineral metabolism in blacks with chronic kidney disease. *Kidney Int*. 2008;73:956–962.
270. Levin A, Bakris GL, Molitch M, et al. Prevalence of abnormal serum vitamin D, PTH, calcium, and phosphorus in patients with chronic kidney disease: results of the study to evaluate early kidney disease. *Kidney Int*. 2007;71:31–38.
271. National Kidney Foundation. K/DOQI clinical practice guidelines for bone metabolism and disease in chronic kidney disease. *Am J Kidney Dis*. 2003;42(suppl 3):S1–S201.
272. Souberbielle JC, Boutten A, Carlier MC, et al. Inter-method variability in PTH measurement: implication for the care of CKD patients. *Kidney Int*. 2006;70:345–350.
273. Kendrick J, Andrews E, You Z, et al. Cholecalciferol, calcitriol, and vascular function in CKD: a randomized, Double-Blind trial. *Clin J Am Soc Nephrol*. 2017;12(9):1438–1446. Epub 2017 Aug 7.
274. Levin A, Tang M, Perry T, et al. Randomized controlled trial for the effect of vitamin D supplementation on vascular stiffness in CKD. *Clin J Am Soc Nephrol*. 2017;12(9):1447–1460. Epub 2017 May 26.
275. Kovesdy CP, Ahmadzadeh S, Anderson JE, et al. Association of activated vitamin D treatment and mortality in chronic kidney disease. *Arch Intern Med*. 2008;168:397–403.
276. Shoben AB, Rudser KD, de Boer IH, et al. Association of oral calcitriol with improved survival in nondialyzed CKD. *J Am Soc Nephrol*. 2008;19:1613–1619.
277. Naves-Diaz M, Alvarez-Hernandez D, Passlick-Deetjen J, et al. Oral active vitamin D is associated with improved survival in hemodialysis patients. *Kidney Int*. 2008;74:1070–1078.
278. Teng M, Wolf M, Ofsthun MN, et al. Activated injectable vitamin D and hemodialysis survival: a historical cohort study. *J Am Soc Nephrol*. 2005;16:1115–1125.
279. Duranton F, Rodriguez-Ortiz ME, Duny Y, et al. Vitamin D treatment and mortality in chronic kidney disease: a systematic review and meta-analysis. *Am J Nephrol*. 2013;37:239–248.
280. Lu RJ, Zhu SM, Tang FL, et al. Effects of vitamin D or its analogues on the mortality of patients with chronic kidney disease: an updated systematic review and meta-analysis. *Eur J Clin Nutr*. 2017;71(6):683–693.

BOARD REVIEW QUESTIONS

- Which of the following are common factors in the uremic phenotype that have been proposed to mediate/contribute to insulin resistance?
 - Dyslipidemia and increased fat mass
 - Protein-energy wasting and inflammation
 - Anemia and acidosis
 - Vitamin D deficiency and hyperparathyroidism
 - a, b, and c
 - All options

Answer: f
- The following comorbidities and medications may contribute to the high prevalence of hypogonadism in male patients with end-stage kidney disease (ESKD) EXCEPT:
 - Obesity and diabetes mellitus
 - Hypertension and cardiovascular disease
 - Protein-energy wasting
 - Medications such as beta blockers and paricalcitol
 - Medications such as angiotensin-converting enzyme inhibitors, angiotensin receptor blockers and cinacalcet

Answer: d
- Which of the following is/are possible adverse reaction(s) to growth hormone (GH) treatment in children?
 - Fluid retention
 - Intracranial hypertension
 - Hypoglycemia
 - Hyperglycemia
 - a, b, and c
 - a, b, and d

Answer: f
- Which of the following is NOT a medical situation linked to vitamin D deficiency?
 - Growth retardation
 - Osteoporosis and fractures
 - Thyroidal dysfunction
 - Left ventricular dysfunction
 - Increased susceptibility to cancer and autoimmune disease

Answer: c
- Which of the following causes disturbed thyroid hormone profile in patients with chronic kidney disease?
 - Malnutrition and inflammation
 - Iodine retention
 - Metabolic acidosis
 - Anemia
 - b and c
 - All options

Answer: f